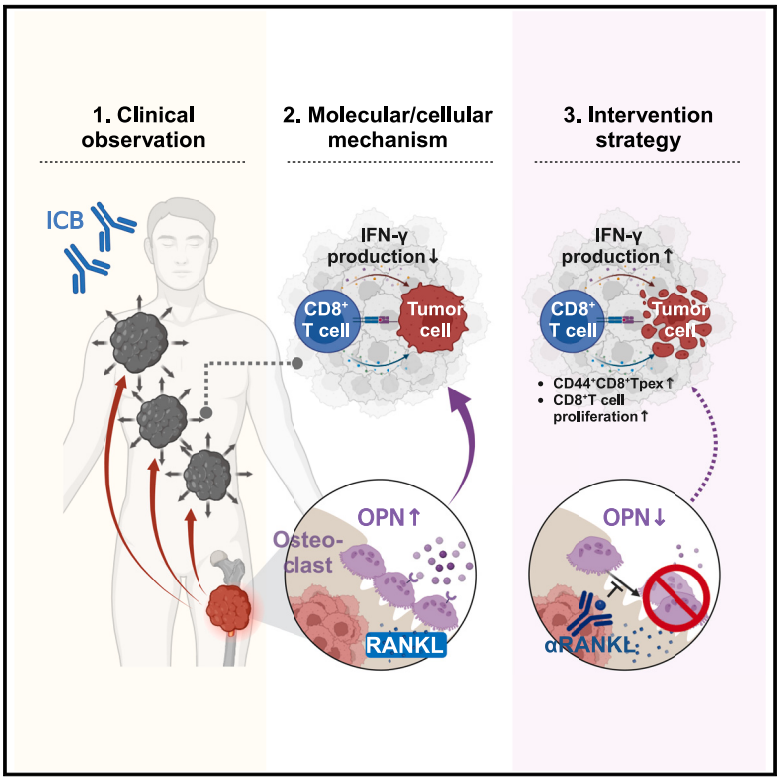
Cancer Cell

Article

Bone metastases diminish extraosseous response to checkpoint blockade immunotherapy through osteopontin-producing osteoclasts

Graphical abstract



Highlights

- Bone metastases drive extraosseous resistance to immune checkpoint blockade (ICB)
- Osteopontin (OPN) is a circulating messenger for immune suppression in primary tumors
- OPN, produced by metastasis-conditioned osteoclasts, impairs T_{pex} in primary tumors
- Targeting osteoclasts or OPN effectively restores sensitivity to ICB therapy

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In brief

Cancer patients with bone metastases often exhibit resistance to immunotherapy. Cheng et al. discover that osseous tumors condition osteoclasts to produce osteopontin, which enters circulation and remotely suppresses precursor exhausted T cells within extraosseous tumors. Blocking RANKL attenuates this immunosuppressive communication, thereby enhancing patient responses to ICB therapy.

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Article

Bone metastases diminish extraosseous response to checkpoint blockade immunotherapy through osteopontin-producing osteoclasts

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SUMMARY

Bone metastatic lesions typically associate with suboptimal responses to immune checkpoint blockade (ICB) therapies. In this study, we observed that across multiple clinical cohorts and a variety of mouse models, the presence of osseous metastases induces ICB resistance in extraosseous tumors. Mechanistically, this long-distance communication is mediated by osseous tumor-conditioned osteoclasts producing osteopontin (OPN). Through circulation, OPN reprograms the extraosseous tumor microenvironment and impairs T cell recruitment and differentiation of $CD8^+TCF1^+$ precursor cells, an essential population for ICB efficacy. In mice, ICB responsiveness is restored by αRANKL blockade of osteoclastogenesis, neutralization of OPN in circulation, or tissue-specific depletion of OPN in osteoclasts. Both the mode of action and therapeutic benefit were validated in clinical cohorts with the αRANKL-ICB combinatory regimen. These findings establish bone as a specific immunoregulatory organ exploited by tumor metastasis and suggest osteoclastogenesis as a promising target to improve ICB prognosis in patients with bone metastasis.

INTRODUCTION

Therapeutic advances in immune checkpoint blockade (ICB) therapies have significantly improved cancer immunotherapy, particularly for solid tumors. However, durable survival benefit remains limited to a minority of patients.[1,2] It has been the focus of the field to identify mechanistic-driven biomarkers to predict patient outcome, and several prognosis biomarkers were

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generated to reflect tumor-intrinsic resistance,[3,4] tumor type-specific[5,6] or mutation-specific[7,8] tumor immune microenvironment (TIME), or systemic impact from comorbidity.[9] However, even in biomarker-stratified patient populations, mostly based on primary tumor analyses, responsiveness to ICB treatment remains generally unsatisfactory. Thus, cancer treatment should shift from focusing solely on the primary tumor to viewing the body as an integrated immune system.

A recent development is the recognition that the organ to which a tumor metastasizes significantly influences ICB treatment outcomes. Tumor cells actively adapt to the microenvironment of different organs, resulting in heterogeneity in the malignant phenotype and treatment responsiveness.[10] For example, liver metastases are more aggressive and resistant to therapy, while metastases to lymph nodes often show stronger responses to ICB.[11–14] Studies focusing on impacts associated with metastasis to organs have provided insights into the immune-related mechanisms responsible for disparities in ICB treatment response.[15–17] Therefore, the concept of organ-specific regulation is increasingly appreciated in the field of cancer immunology, especially as it relates to ICB immunotherapy.[18]

Bone is a common site of metastasis, particularly in breast, prostate, and lung cancers.[19] Clinical trials have linked bone metastases to poor clinical outcomes in ICB-treated patients.[20,21] Several studies have characterized a distinct tumor microenvironment (TME) in bone metastatic lesions.[22,23] Specifically, TGF-β released during bone resorption inhibits Th1 differentiation while promoting Th17 polarization, contributing to ICB resistance.[22] Bone metastases are enriched with inflammatory monocytes producing CCL20, which impairs $CD8^+$ T cell function. Disrupting the CCL20-CCR6 axis has been shown to reduce T cell exhaustion and extend survival after ICB treatment.[23] Notably, recent work has revealed that both extraosseous tumors[24,25] and bone metastasis[26] may alter hematopoiesis in the bone to favor immune tolerance, extending the scope of osteo-immune interplay to extraosseous sites.

As the pivotal site of hematopoiesis, osteoclasts, osteoblasts, and osteocytes shape an osteo-immune environment. In both physiological homeostasis and pathological inflammation, these cells respond to stimuli and regulate the hematopoiesis outcome.[25] In this study, we explored the interplay between bone metastases and extraosseous tumor response to immunotherapy.

RESULTS

Bone metastasis predicts unfavorable therapeutic outcomes and reduced extraosseous tumor responses to ICB

To investigate how metastases in different organs affect the responsiveness to ICB immunotherapy, we first examined the metastatic organ-specific survival in two independent pan-cancer cohorts of patients with metastatic solid tumors who received ICB treatment: AMU1 (n = 927) and AMU2 (n = 343) (Table S1). We observed that progression-free survival (PFS) varied according to the metastatic organ (Figures S1A and S1B). While the remote immunosuppression of liver metastasis was reported previously,[26] we also observed a strong prognosis differentiation made by bone metastasis in comparison to the cohort average (AMU1, $^{**}p = 0.003$; AMU2, $^{**}p = 0.009$, Figures S1A and S1B), or by grouping patients based on the presence or absence of bone metastasis (Figures 1A and 1B).

We extended our analysis to single tumor type cohorts: TUSM1, 2, and 3 (Table S1). All analyses confirmed that the presence of bone metastasis was consistently associated with faster progression in ICB-containing treatments (Figures 1C–1E). Further grouping patients based on primary tumor types, we identified a widely established correlation between bone metastasis and poor responsiveness to ICB therapies (Figure S1C). Stratified analysis showed that bone metastasis is generally correlated with decreased PFS regardless of age, gender, and treatment regimens (Figures S1D–S1F). Multivariate Cox proportional regression analysis further confirms bone metastasis as an independent predictor of unfavorable PFS to ICB therapies (Figures 1F–1H). Notably, such correlation was not observed in bone metastasis patients receiving chemotherapy alone either in the pan-cancer cohort (AMU3, n = 81), NSCLC cohort (AMU4, n = 237) or a small cell lung cancer cohort (SCLC) (AMU5, n = 71) (Figures S1G–S1J). This suggests that bone metastasis stratifying patients' outcomes is in the context of immunotherapy.

We further evaluated the bone metastasis impact on ICB in overall survival (OS) data-available cohorts: TUSM2, TUSM3, and Nat Med 2019 (a published melanoma cohort with 144 patients),[27] and observed a clear separation of patients' outcomes (Figures 1I–1K). Generic inverse variance meta-analysis also confirmed that baseline bone metastasis was consistently associated with shorter survival across multiple cohorts (Figure 1L).

We next analyzed objective response rate (ORR), a standard measurement of tumor shrinkage according to RECIST 1.1 criteria.[28] Objective response reflects the change in the sum of the diameters of measurable lesions, but bone lesions are almost unmeasurable. Therefore, the ORR in patients with bone metastasis largely reflects responses in extraosseous lesions. Across all five ORR data-available cohorts, we observed that baseline bone metastasis is correlated with lower ORR to ICB therapies (Figure 1M). A pooled analysis also confirmed that the ORR to ICB therapies were significantly lower in patients with bone metastasis (Figure 1N).

The extent of tumor shrinkage could serve as a proxy for long-term treatment outcomes against solid tumors.[29] We focused on the burden of extraosseous lesions in the AMU6 (pan-cancer, n = 40) and AMU7 (pan-cancer, n = 59) cohorts to directly assess the long distance impact of bone metastasis (Figure 2A; Table S1). In AMU6 cohort, we found that the extraosseous tumor burden was significantly reduced after ICB treatment in patients without bone metastasis, whereas no significant tumor shrinkage was observed in ICB-treated bone metastatic patients (Figure 2B). Accordingly, patients with bone metastasis who received ICB treatment showed substantially less changes in extraosseous tumor burden than those without bone metastasis (Figure 2C). This reduced shrinkage of extraosseous tumors was also observed in the larger sample size of AMU7 cohort (Figures 2D and 2E).

To evaluate whether the impact of bone metastasis on extraosseous tumor response to ICB therapies is organ-specific, we reviewed radiological follow-up data for patients with lung as the primary tumor site in cohort TUSM2. In this 48 NSCLC patient cohort, we found significantly less tumor shrinkage in lesions

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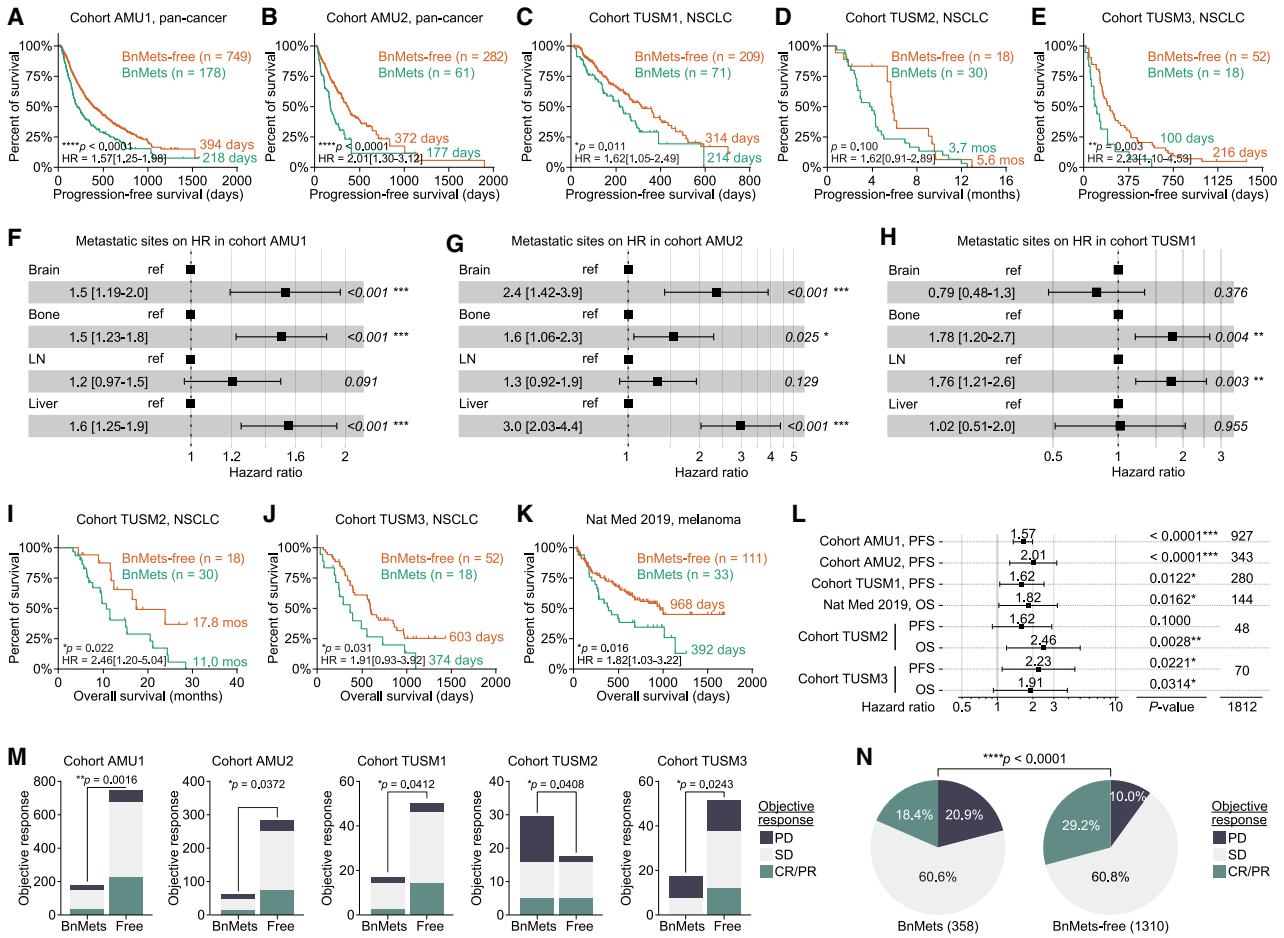


Figure 1. Bone metastasis on therapeutic response to ICB immunotherapy

(A–E) Bone metastasis (BnMets) on PFS for cohort (A) AMU1 (n = 927), (B) AMU2 (n = 343), (C) TUSM1 (n = 280), (D) TUMS2 (n = 48), and (E) TUSM3 (n = 70).

(F–H) Predictive significance of metastatic site on PFS for cohort (F) AMU1, (G) AMU2 and (H) TUSM1.

(I–K) BnMets on OS for cohort (I) TUSM2, (J) TUSM3, and (K) Nat Med 2019, melanoma (n = 144).

(L) Forest plot showing the hazard ratios in patients who received ICB treatment, stratified by the BnMets.

(M) Barplots showing the objective response to ICB treatment, stratified by the BnMets.

(N) Pie charts showing the responses of all patients in cohorts represented in (M). Statistical analysis: Log rank test (A-E, I-L), cox proportional hazard regression analysis (F-H), and Chi-square test (M, N) was used. p-values <0.05 were considered significant. See also Figure S1 and Table S1.

located in the lungs as primary lesions (p = 0.050), intrapulmonary metastasis (*p = 0.023), livers (*p = 0.042), lymph nodes (*p = 0.030), and pleural nodules (*p = 0.029) (Figure 2F). Collectively, these findings suggest that bone metastasis may lead to unfavorable therapeutic outcomes in patients receiving ICB immunotherapy through a systemic impact on extraosseous tumors.

The presence of osseous tumors diminishes extraosseous αPD-L1 response in mice

To examine the causality between the bone metastases and extraosseous resistance to ICB therapies, we set up a mouse model of bone metastasis (Figure S2A). B16F10-luciferase tumor cells were injected into the proximal tibia cavity through a plateau (intratibial injection; designated as IT mice).[22] *In vivo* bioluminescence imaging (BLI) confirmed the formation of osseous tumor colonies at 10 days post transplantation in 100% of tested animals (Figure S2B). Micro-computed tomography revealed typical radiological manifestations of bone metastasis in our model: defects developed in cortical bones followed by erosive bone remodeling (Figures S2C–S2F). H&E staining confirmed the establishment of osseous lesions (Figures S2G and S2H).

Simultaneously, tumor cells (B16F10 in Figure 3A) were implanted in IT mice subcutaneously as an extraosseous tumor (designated as SC + IT mice). Subcutaneous tumor-only mice (designated as SC mice) serve as controls. BLI showed that compared to the isotype control group αPD-L1 treatment resulted in a significant reduction of tumor burden in SC control mice (Figure 3B). The therapeutic effect was not observed for osseous tumors in SC + IT mice, consistent with reported evaluation of osseous tumor burden (Figure 3C).[22] Notably, subcutaneous tumors in SC + IT mice also showed no significant response to αPD-L1 treatment, recapitulating extraosseous resistance to ICB therapies in patients (Figure 3D). Saline

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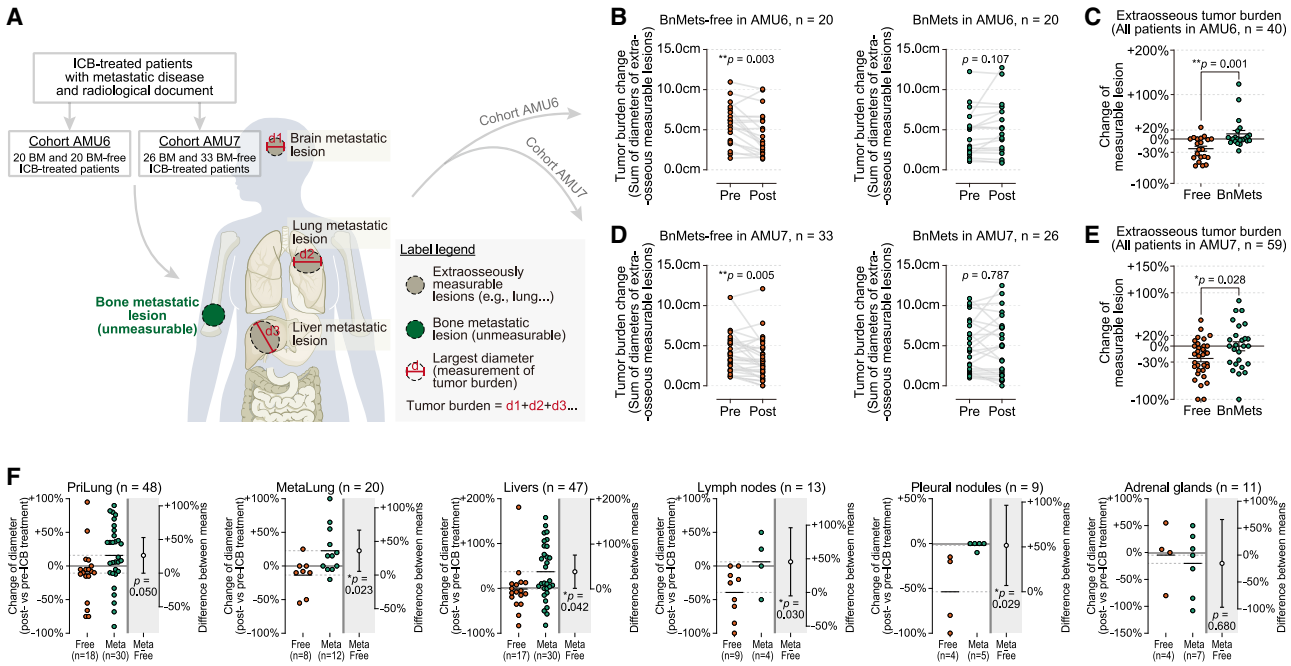


Figure 2. Inferior extraosseous tumor shrinkage following ICB treatment in patients with bone metastasis

(A) Workflow for evaluating extraosseous tumor shrinkage after ICB administration. BM, bone metastasis.

(B) Extraosseous tumor burden before and after ICB administration for cohort AMU6 (n = 40).

(C) Extraosseous tumor burden between patients with BnMets-free and BnMets disease for cohort AMU6 (n = 40).

(D) Extraosseous tumor burden before and after ICB administration for cohort AMU7 (n = 59).

(E) Extraosseous tumor burden between patients with BnMets-free and BnMets disease for cohort AMU7 (n = 59).

(F) Changes in the tumor burden of extraosseously measurable tumors in ICB-treated patients in cohort TUSM2. Dots indicated the changes for each patient. Differences between the means are presented in gray shadow. PriLung, primary lung lesion; MetaLung, intrapulmonary metastatic lesion. Statistical analysis: Paired two-tailed Student's t test (B, D), and unpaired two-tailed Student's t test (C, E, F) was used. p-values <0.05 were considered significant. See also Table S1.

injection confirmed that this extraosseous resistance is not caused by the operational stress from IT injection (Figure S3A).

This extraosseous αPD-L1 resistance was also observed in B16F10-OVA, MC38, and CT26 colorectal carcinoma, and LLC-OVA NSCLC models (Figures 3E, 3F, 3H and S3B–S3D). Consistently, shortened survival was observed in all aforementioned models (Figures 3G and 3H and S3E–S3G). We also studied extraosseous ICB resistance in lung metastasis. In the pulmonary metastasis model, B16F10-OVA cells were injected intravenously (designated as IV + IT mice, Figure 3I). While nodule formation was significantly reduced upon αPD-L1 treatment in osseous tumor-free IV control littermates, pulmonary nodule formation showed no obvious response in αPD-L1-treated IV + IT mice (Figure 3J).

To rule out the possibility that the diminished αPD-L1 response in extraosseous tumors is purely due to increased tumor burden from forming osseous lesions, we compared mice with unilateral subcutaneous MC38 tumors to those implanted bilaterally (Figure 3K). In the bilateral implantation group, we adjusted the cell number so that the pre-treatment subcutaneous tumors reached a similar burden as the intratibially injected osseous tumors (Figures 3L and 3M). Upon treatment with the αPD-L1 antibody, we observed that both tumors implanted bilaterally responded efficiently, similar to unilaterally implanted ones. Additionally, doubling the baseline number of tumor cells did not impair the treatment efficacy (Figure 3N). Taken together, we reasoned that the suppression of extraosseous responses to ICB by osseous tumors is not due to the increased tumor burden.

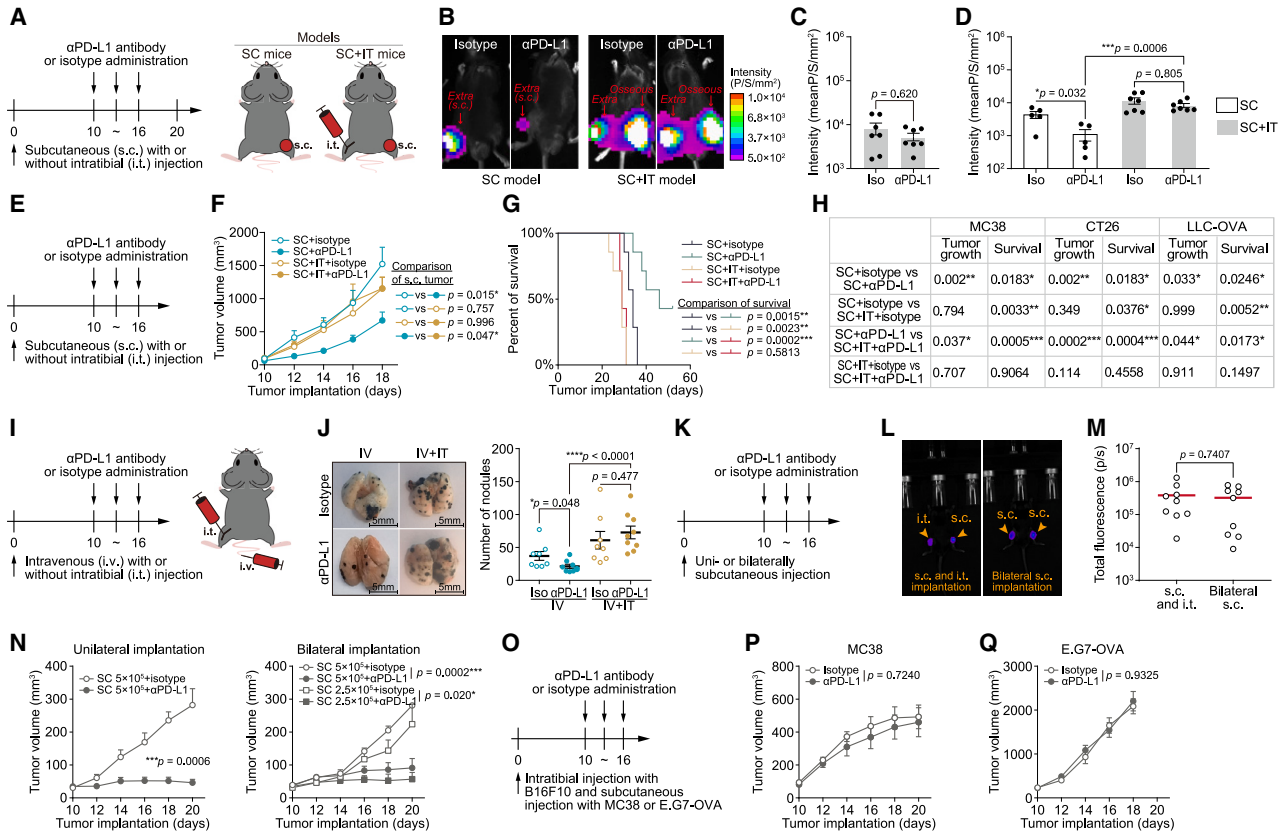
Next, to investigate whether osseous tumor-associated ICB resistance is tumor antigen-specific, in mice bearing osseous B16F10 tumors, we subcutaneously implanted either MC38 or E.G7-OVA tumors, both of which are highly sensitive to ICB (Figure 3O). The subcutaneous tumors, whether MC38 (Figure 3P) or E.G7-OVA (Figure 3Q), resisted αPD-L1 treatment. This observation suggests that the immunosuppression generated by osseous tumors is unlikely limited to osseous tumor-specific antigen, but rather affects a much broader range of anti-tumor immunity.

Osseous lesions impair CD8$^+$ T cell immunity in extraosseous tumor

To investigate the effects of bone metastasis on extraosseous tumors, we harvested subcutaneous tumors from SC or SC + IT mice 10 days post tumor implantation for RNA-seq. Our Immune Map analysis[30] showed that, prior to any treatment, the bone lesions generated a "cold" TIME at baseline, indicating an unfavorable TIME to ICB therapies (Figure 4A). Specifically, focusing on a well-established 18-gene signature predicting clinical ICB responsiveness,[31] we observed a broad suppression of expression (Figure 4B). Gene set enrichment analysis

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	MC38		CT26		LLC-OVA	
	Tumor growth	Survival	Tumor growth	Survival	Tumor growth	Survival
SC+isotype vs SC+αPD-L1	0.002**	0.0183*	0.002**	0.0183*	0.033*	0.0246*
SC+isotype vs SC+IT+isotype	0.794	0.0033**	0.349	0.0376*	0.999	0.0052**
SC+αPD-L1 vs SC+IT+αPD-L1	0.037*	0.0005***	0.0002***	0.0004***	0.044*	0.0173*
SC+IT+isotype vs SC+IT+αPD-L1	0.707	0.9064	0.114	0.4558	0.911	0.1497

Figure 3. Osseous tumors diminish the extraosseous response to αPD-L1 treatment in mouse model

(A) Schematic for establishing mouse model bearing both osseous and extraosseous tumor.

(B) Representative images of bioluminescence imaging (BLI) at 20 days after tumor implantation.

(C and D) Intensity of bioluminescence from (C) osseous ($n = 7$ for each group) and (D) extraosseous tumors in SC ($n = 5$ for each group) and SC + IT mice ($n = 7$ for each group) treated with either αPD-L1 or isotype.

(E) Schematic for tumor implantation and αPD-L1 treatment.

(F) Growth curves of s.c. B16F10-OVA tumors in SC + IT mice treated with either αPD-L1 ($n = 6$ for SC and $n = 7$ for SC + IT) or IgG isotype ($n = 6$ for SC and $n = 7$ for SC + IT).

(G) Survival curves of αPD-L1 or isotype-treated SC or SC + IT B16F10-OVA-bearing mice ($n = 7$ for each group).

(H) Summary of the p-values for the statistical analyses comparing tumor growth and survival from the data shown in Figures S3B–S3G.

(I) Schematic for tumor implantation and αPD-L1 treatment.

(J) Representative lungs (left panel) and summary of nodule formation (right panel) in IV ($n = 8$ for each group) or IV + IT implanted ($n = 8$ for Iso and $n = 9$ for αPD-L1) B16F10-OVA tumor-bearing mice, either treated with αPD-L1 or IgG isotype. Two-tailed Student's t test was used.

(K) Strategy for evaluating the effect of tumor burden on the efficacy of αPD-L1 treatment.

(L) Representative images and (M) total bioluminescence levels ($n = 9$ for each group) at 10 days after tumor implantation. Red line represents mean value.

(N) Tumor growth in unilaterally ($n = 7$ for each group) and bilaterally subcutaneously implanted MC38 tumor-bearing mice, treated with either αPD-L1 ($n = 6$ for each group) or isotype ($n = 5$ for each group).

(O) Schematic for tumor implantation and αPD-L1 treatment.

(P and Q) Growth curves in osseous tumor-bearing mice with subcutaneously implanted tumors, derived from (P) MC38 ($n = 7$ for Iso and $n = 6$ for αPD-L1) or (Q) E.G7-OVA ($n = 7$ for each group), that were treated with either αPD-L1 or isotype. Three or more independent experiments were performed, and the data shown are from a single experiment that is representative. Statistical analysis: Mean ± SEM (C, D, J, M) are shown. Two-tailed Student's t test (C, D, J, M), two-way ANOVA (left panel in N, P, Q) and two-way ANOVA with Tukey's post hoc test (right panel in N, F, H), and log rank test (G, H) was used. p-values <0.05 were considered significant. See also Figures S2 and S3.

(GSEA) against three clinically validated ICB-responsive signature—CheckMate-275,[32] POPLAR,[33] and KEYNOTE-028[34] —confirmed that the extraosseous TIME from SC + IT mice is less sensitive for ICB responses (Figure 4C). GSEA further revealed that the most significant suppression of the baseline transcriptome involved genes related to pro-inflammatory responses, such as IL-2-STAT5 signaling and interferon-α/γ responses, which are critical for the responsiveness to ICB therapies in extraosseous tumors (Figure 4D).

To depict osseous lesion-induced suppression of the extraosseous TIME at the cellular level, we employed mass cytometry by time-of-flight (CyTOF) and multi-dimensional scaling (MDS) analysis, which separated the subcutaneous tumors between groups with or without IT implantation (Figure 4E). CD45$^+$

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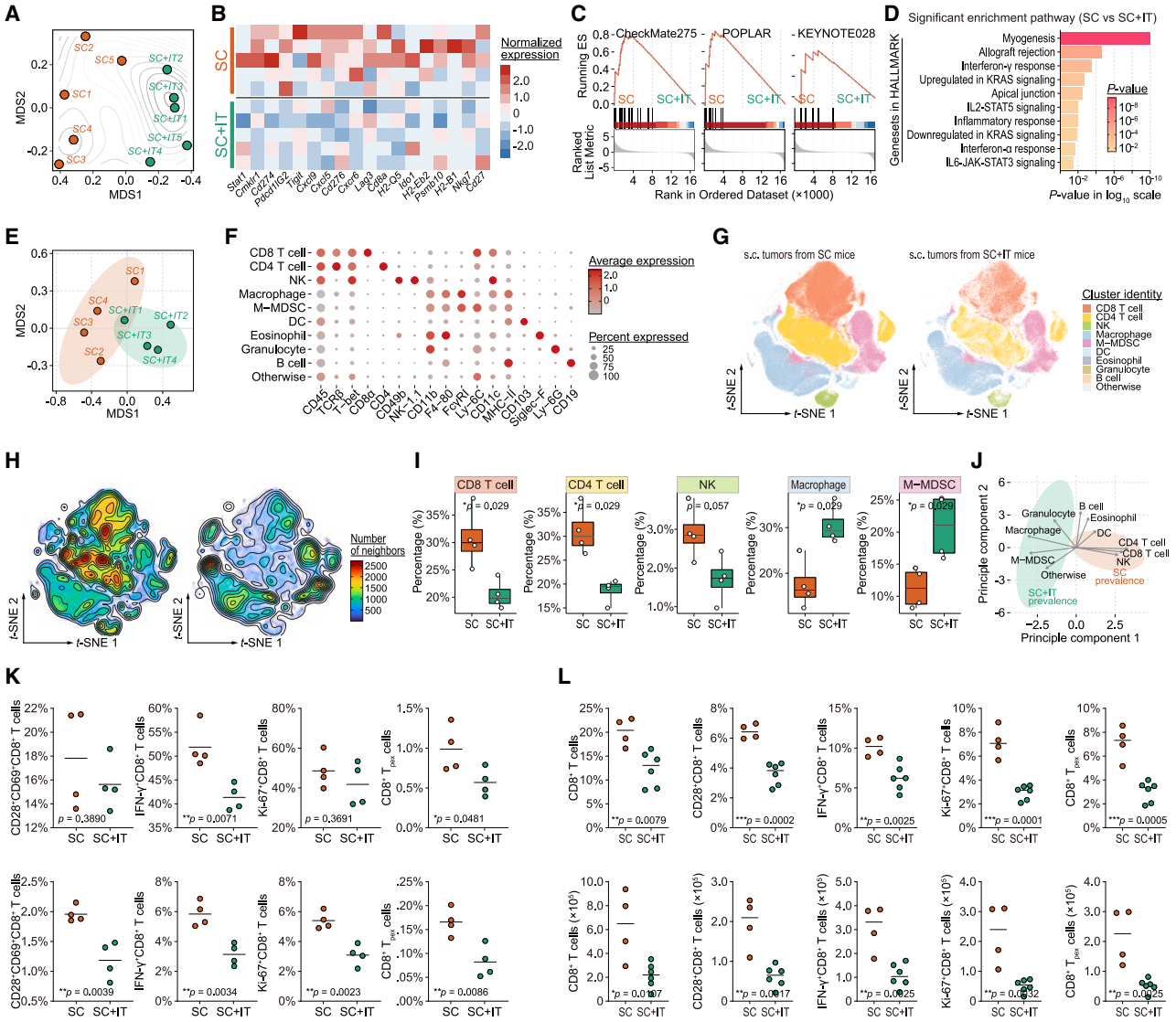


Figure 4. Bone metastasis altered the extraosseous immune microenvironment

(A) Scatterplot showing the immune map of subcutaneous tumor from SC and SC + IT B16F10 tumor-bearing mice (n = 5 for each group).

(B) Row-scaled normalized expression of genes in IFN-γ-related pathway.

(C) GSEA showing expression of genes in the ICB-responsive signature in s.c. tumors.

(D) Enrichment of different signatures in HALLMARK collection.

(E) MDS analysis of replicate samples (n = 4 for each group).

(F) Row-scaled expression of feature proteins.

(G) t-SNE projections of the detected cells, colored by annotated population (n = 4 for each group).

(H) t-SNE clustering were colored according to the cell density.

(I) Immune cell infiltration in SC and SC + IT mice. Bottom, middle and upper lines of boxplot denote lower quartile, median and upper quartile, respectively. Whiskers denote range.

(J) Biplot showing contributing immune cell types identified via principal component analysis (PCA).

(K and L) Percent composition of each cell type in s.c. tumors. Horizontal line represents mean value. (K) Clustering of $CD8^+$ T cells in CyTOF analysis. (L) $CD8^+$ T cells phenotype in MC38-OVA tumor-bearing mice by FACS (n = 4 for SC and n = 6 for SC + IT). Three or more independent experiments were performed, and the data shown are from a single experiment that is representative. Statistical analysis: Two-tailed Student's t test (K, L), and Mann-Whitney U-test (I) was used. p-values <0.05 were considered significant.

cells were clustered in an unsupervised manner into 10 populations (Figure 4F and 4G). Overall, the TIME of SC mice was enriched with cell populations associated with anti-tumor immunity, such as $CD8^+$ T cells, $CD4^+$ T cells, and NK cells, whereas pro-tumor immunoregulatory cells, such as macrophages and M-MDSCs, were augmented in the TIME of SC + IT groups (Figures 4H and 4I). Principal component analysis (PCA) further validated this bifurcational polarization of

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lymphoid versus myeloid enrichments in the SC versus SC + IT TIME, respectively (Figure 4J).

With the CyTOF data, we further analyzed the $CD8^+$ T cell phenotypes. Within the $CD8^+$ population, we detected that IFN-γ-producing cells and progenitor exhausted T cells (T_{pex}, $CD8^+CD44^+PD\text{-}1^+TCF1^+TIM\text{-}3^-$) were reduced in the SC + IT TIME (Figure 4K). It has been well-documented that T_{pex} cells are the dominant population responding to ICB treatment for proliferation and prolonged anti-tumor efficacy.[35,36] Since the overall infiltration of $CD8^+$ T cells was suppressed in the SC + IT TIME, other subgroups, including early reactivated ($CD28^+CD69^+$) and proliferating (Ki-67^+) $CD8^+$ T cells, were also reduced in absolute numbers (Figure 4K). All these observations were subsequently confirmed by independent experiments with flow cytometry analyses (Figure 4L). These data suggest that osseous lesions remotely reprogram the extraosseous TIME to diminish $CD8^+$ T cell infiltration and effector differentiation necessary for ICB responses.

Osteoclasts contribute to extraosseous ICB resistance

We hypothesized that the reprogramming of the extraosseous TIME occurs through remote communication with the osseous lesion. To test this hypothesis, we collected serum from tumor-free, SC, or IT mice donors and subsequently transferred it to recipient mice with subcutaneous tumors only. αPD-L1 antibody or the isotype control was concurrently administered to the serum recipients (Figure 5A). Sera collected from IT donors, but not from tumor-free or SC donors, recapitulated the bone lesion effect in diminishing subcutaneous tumors' response toward αPD-L1 treatment, in both B16F10-OVA (Figure 5B) and MC38 (Figure 5C) tumor model. Furthermore, we transferred sera from donors carrying B16F10-OVA bone tumors to MC38-bearing recipients (Figure 5D). Sera from mice carrying unmatched bone tumors also significantly attenuated the MC38 tumor's response to αPD-L1 (Figure 5E), which was consistent with observed ICB resistance in s.c. tumors derived from unmatched osseous lesions (Figures 3O–3Q).

To identify signals sent to extraosseous tumors that attenuate ICB responsiveness, we performed quantitative proteomic profiling on serum from IT, SC, and tumor-free mice. PCA analysis unbiasedly grouped the sera from these sources into three distinct clusters (Figure 5F and Table S2). In comparison to sera from SC mice, despite both donors carrying the same type of tumor, the osseous location of the tumors reshaped the sera protein spectrum by upregulating 161 proteins and downregulating 163 proteins (Figure 5G and 5H and S4A–S4F).

In this dataset, we observed an enrichment in IT mice sera of proteins promoting bone remodeling (Figure S4F). Bone remodeling is a hallmark of established bone metastasis,[37,38] and coordinated by the activity between osteoblast and osteoclast.[39] In mice bearing osseous or subcutaneous tumors, the difference in serum composition tips the balance between osteoblast and osteoclast: an over-representation of osteoclastogenesis-promoting molecules such as FBN1 and RHOA in the sera from IT mice, and an over-representation of osteoblast-favoring molecules such as CLIC1, LTF, and IGF1 in the SC sera (Figure 5I). This serum profile predicts bone loss. We evaluated osseous tumor-related bone disruption *in vivo* with a fluorescent bisphosphonate derivative, OsteoSense 750EX, to label active bone mineralization. Fluorescence molecular tomography analysis of the tibial plateau showed substantially reduced hydroxyapatite deposits in limbs implanted with B16F10-OVA (Figure 5J) or MC38 (Figure S4G) tumors. The heightened osteoclast activity was also reflected by the substantial loss of bone mineral density in the tumor-bearing limbs (Figures S4H and S4I).

To establish the contribution of osteoclast to the ICB resistance of extraosseous tumors, as previously described,[40,41] we differentiated and matured bone marrow mononuclear cells into osteoclasts in culture. The osteoclast-conditioned medium was collected and intraperitoneally administered to subcutaneous tumor-bearing recipients (Figure 5K). TRAP staining confirmed the functional maturation of cultured osteoclasts (Figure 5L). Analysis of subcutaneous tumor progression showed that osteoclast-conditioned medium largely blunted responsiveness to αPD-L1 blockade in both B16F10-OVA (Figure 5M) or MC38 (Figure 5N) tumor-bearing recipients. These data collectively suggested that the signal that remotely reprograms the ICB responses of extraosseous tumors may be generated by osteoclasts.

Overproduction of osteopontin by tumor-conditioned osteoclasts induces ICB resistance

Phenotypic plasticity of osteoclasts is well-documented,[42] prompting us to investigate whether osteoclasts have unique traits in the presence of osseous tumors. To determine the effects of tumors and avoid the confounding influence of other environmental factors *in vivo*, we employed a reductionist approach to characterize *in vitro* differentiated osteoclasts in the absence or presence of tumor-conditioned culture medium (Figure 6A). RNA-seq analysis revealed that tumor-conditioning substantially reprogramed the osteoclast transcriptome, with significant upregulation of 3,467 genes (Figure 6B). Intersecting this gene set with the serum proteomics dataset identified 19 overlapping candidates whose mRNAs were upregulated in the tumor-conditioned osteoclast culture and whose proteins were over-represented in the serum of bone tumor-carrying mice (Figure 6C). Among them, *Spp1*, which encodes osteopontin (OPN) and specifically upregulated in IT serum (Figure 6D), is the only secreted protein and has been previously reported to reprogram the TME.[43,44]

We confirmed that adding tumor cell-conditioned medium (TCM) significantly upregulated *Spp1* mRNA and effectively increased OPN protein secretion in both bone marrow-derived osteoclasts and isolated primary osteoclasts (Figure 6E and 6F and S5A and S5B). Additionally, we confirmed that establishing intratibial tumors elevated OPN concentration in sera (Figure 6G). It has been previously reported that deletions of the *Spp1* gene in both tumor and recipient mice leads to intratumorally immune activation.[43] GSEA analyses with RNA-seq data collected from subcutaneous tumors accompanied by or without bone lesions (Figure 4A) against a published dataset of OPN-sufficient versus OPN-deficient tumors[43] showed concordant suppression of genes involved in T cell proliferation, IFN-γ production and responses, as well as anti-tumor cytotoxicity (Figure 6H).

To validate the contribution of OPN to extraosseous ICB resistance, we neutralized OPN in serum from IT donor mice using an antibody, then transferred the treated serum to SC recipient mice (Figure 6I). During αPD-L1 treatment, the resistance

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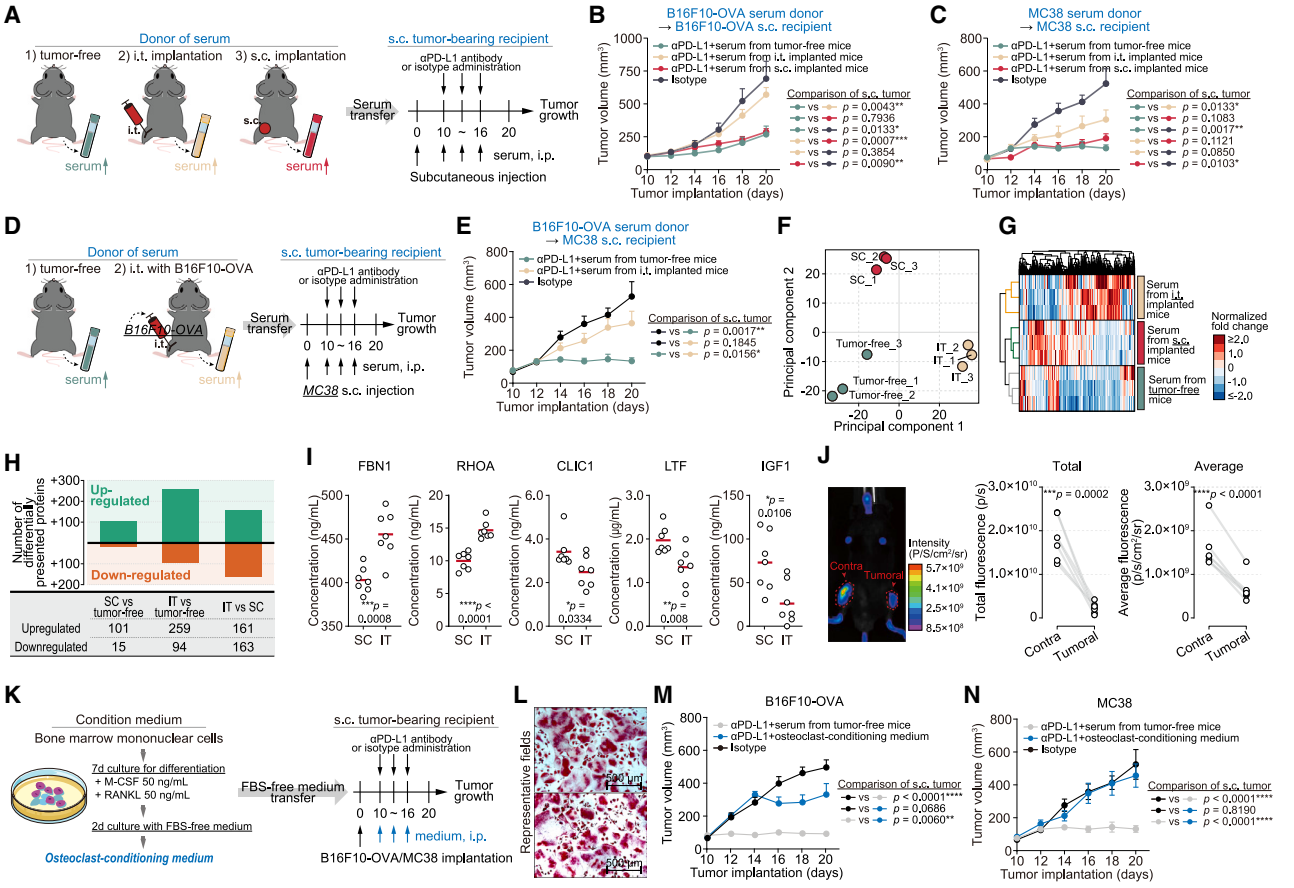


Figure 5. Osteoclasts drive extraosseous ICB resistance

(A) Schematic for tumor implantation, αPD-L1 treatment, and serum transfer.

(B and C) Growth curves in αPD-L1-treated s.c. tumor-bearing recipients. Tumor growth was measured either in (B) B16F10-OVA ($n = 5$ for isotype group and $n = 6$ for every other group) or (C) MC38 recipients ($n = 6$ for tumor-free serum, $n = 5$ for s.c. serum, $n = 7$ for i.t. serum, and $n = 6$ for isotype).

(D) Schematic for tumor implantation, αPD-L1 treatment, and serum transfer.

(E) Growth curves in αPD-L1-treated s.c. MC38 tumor-bearing recipients.

(F) PCA analysis showing the bio-similarity of serum samples from different sources. Dots represent replicates from each source ($n = 3$ for each group).

(G) Relative abundances of differentially expressed proteins.

(H) Number of differentially expressed proteins.

(I) Concentration of interested proteins related to osteoclast or osteoblast ($n = 7$ for each group) by ELISA. Red line represents mean value.

(J) Hydroxyapatite deposits in the regions of interest (ROIs) for B16F10-OVA osseous tumor-bearing mice ($n = 7$ for each group). Contra, tumor-free limb; tumoral, intratibially implanted.

(K) Schematic for tumor implantation, αPD-L1 treatment, and medium transfer.

(L) TRAP staining revealed the differentiated bone marrow mononuclear cell-induced osteoclasts.

(M and N) Growth curves in αPD-L1-treated s.c. tumor-bearing recipients, who were transferred with osteoclast-conditioned medium. Tumor growth was measured either for (M) B16F10-OVA or (N) MC38 ($n = 6$ for tumor-free serum, $n = 7$ for OCT supernatant, and $n = 6$ for isotype). Statistical analysis: two-way ANOVA with Tukey's post hoc test (B, C, E, M, N), unpaired (I) and paired (J) two-tailed Student's *t* test was used. Data in survival curve were shown as mean ± SEM. *p*-values <0.05 were considered significant. See also Figure S4 and Table S2.

generated by OPN-neutralized IT mouse sera was significantly reduced (Figure 6J). Additionally, the direct combination of αOPN neutralizing antibody with αPD-L1 (Figure 6K) significantly sensitized the subcutaneous tumors of mice bearing bone lesions to benefit from ICB treatment (Figure 6L).

OPN can be expressed in a wide range of cells.[45] We investigated whether osteoclasts are the primary source of OPN in serum under bone metastasis conditions, and whether osteoclast-produced OPN alone was sufficient to diminish extraosseous ICB responses. Osteoclast-specific OPN depletion was achieved in mice through cathepsin K (*Ctsk*)-driven *Cre* expression[46] in mice with floxed *Spp1* alleles (Figure 6M). Osteoclast-specific OPN depletion significantly decreased serum OPN concentration at 10 days after SC + IT implantation (prior to αPD-L1 administration), suggesting osteoclast as a major producer of OPN in bone tumor-bearing model (Figure 6N). In comparison to control mice ($Spp1^{f/f}$), extraosseous tumor responses to αPD-L1 in the presence of bone tumor were significantly improved in *Spp1*-deleted mice ($Ctsk^{Spp1}$) (Figure 6O), and survival time length was extended (Figure 6P). FACS analyses of

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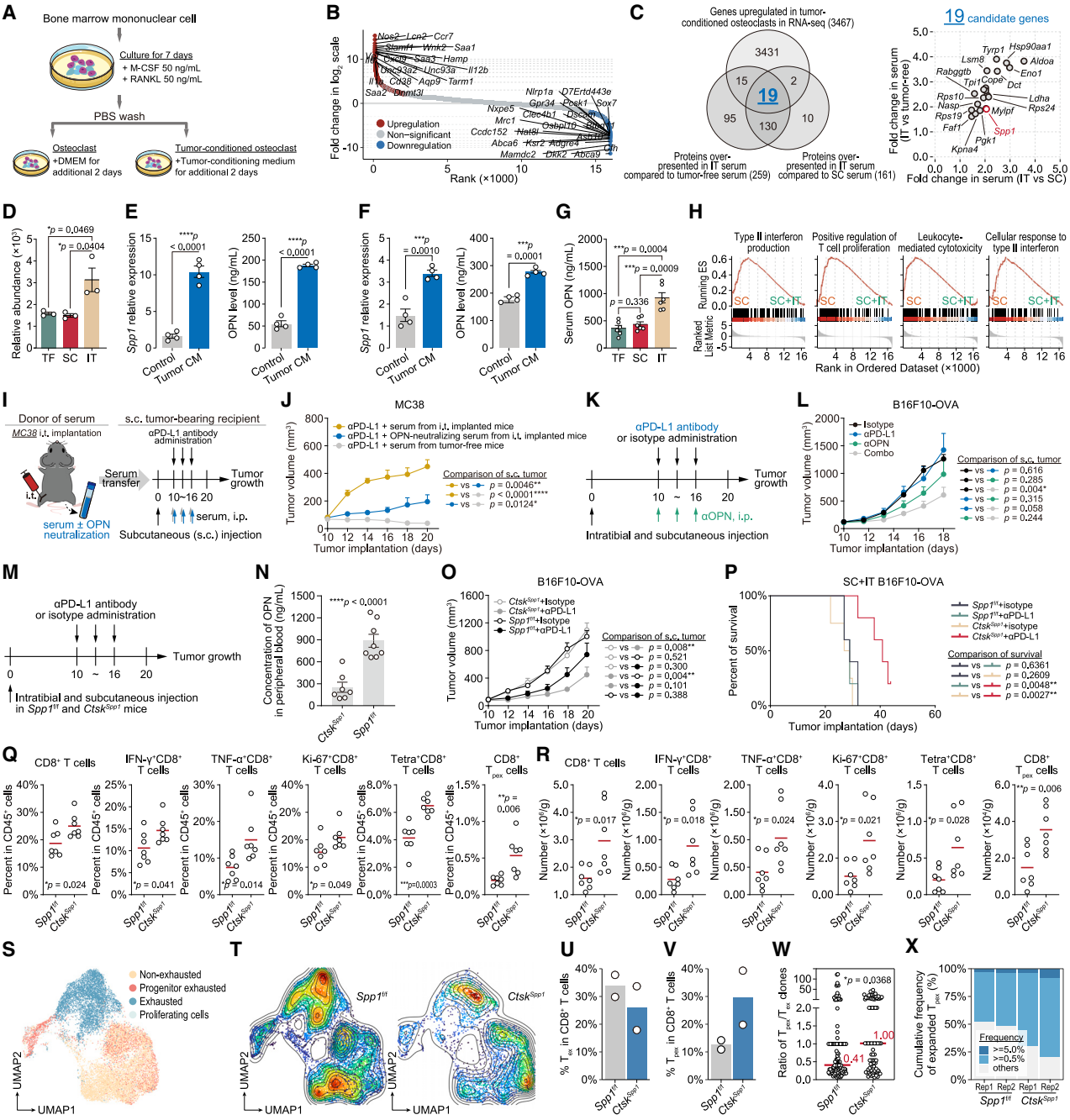


Figure 6. OPN in osteoclast-mediated extraosseous ICB resistance

(A) Strategy for evaluating tumor-induced phenotyping of osteoclasts *in vitro*.

(B) Differentially regulated genes between osteoclasts cultured in the presence of tumor-conditioned or control medium.

(C) Strategy for identifying mediators of extraosseous αPD-L1 resistance.

(D) Relative abundance of OPN by quantitative proteomics (n = 3 for each group). Mean ± SEM.

(E and F) Quantifications of *SPP1*/OPN production from bone marrow-derived osteoclasts (E) and primary osteoclasts (F) after cultured in tumor-conditioned medium (n = 4 for each group). Tumor CM, tumor-conditioned medium. Mean ± SEM.

(G) ELISA assay comparing OPN concentration in serum samples (n = 6 for each group). Mean ± SEM.

(H) Enrichment of OPN-regulated pathway in s.c. tumors. NES, normalized enrichment score.

(I) Strategy for determining the impact of OPN on αPD-L1 treatment.

(J) Growth curves in αPD-L1-treated s.c. tumor-bearing recipients who were transferred with serum from tumor-free mice, IT donor mice, or IT donor mice treated with αOPN neutralizing antibody (n = 6 for each group).

(K) Strategy for evaluating OPN neutralization on extraosseous αPD-L1 resistance.

(legend continued on next page)

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subcutaneous tumors showed an increase in T cell infiltration while reducing the infiltration of immunosuppressive cells such as MDSC and TAM in *Ctsk*^{Spp1} mice, which supports our speculation that osteoclast secretion of OPN inhibits extraosseous anti-tumor immunity (Figure S5C). Focusing on CD8$^+$ T cells, a significantly improved baseline CD8$^+$ T cell infiltration, including those surrogate tumor antigen-specific (Ova-tetramer$^+$) T cells increased in in *Ctsk*^{Spp1} mice. Additionally, we observed heightened CD8$^+$ T cell proliferation, increased IFN-γ and TNF-α-producing effector T cell, as well as enhanced differentiation of T_{pex} cells (Figures 6Q and 6R). Meanwhile, we observed a slight decrease in the splenic MDSC population, which is consistent with the trend observed in subcutaneous tumors (Figure S5D).

To further characterize the features of intratumoral CD8$^+$ T cells, we performed scRNA- and scTCR paired-sequencing with CD8$^+$ T cells collected from treatment-naïve tumors hosted by *Ctsk*^{Spp1} or *Spp1*$^{f/f}$ mice. After filtering and quality control, CD8$^+$ T cells were categorized into four general phenotypes: non-exhausted, progenitor of exhausted (T_{pex}), exhausted (T_{ex}), and proliferating cells (Figures 6S and S6). In agreement with FACS assays, we observed an increase in T_{pex} cells in s.c. tumors from SC + IT *Ctsk*^{Spp1} mice and a reciprocal decrease in T_{ex} in percentage (Figures 6T–6V).

T cell repertoire sequencing allows us to track T cell differentiation at the individual clone level. For all 2,600 clonotypes of TCRs, we compared their relative T_{pex} to T_{ex} ratio. Overall, this ratio increased more than 2-fold upon deletion of the *Spp1* gene in hosts' osteoclasts (Figure 6W). Zooming in on those T_{pex} cells, we found that, upon OPN depletion, the frequency distribution of the T_{pex} repertoire was more dominated by highly expanded clonotypes (frequency $\geq 0.5\%$[47]), showing more efficient clonal expansion of this critical subset determining the ICB efficacy (Figure 6X). Overall, these findings indicated that OPN produced by osteoclasts is the major signal sent by bone lesions to reinforce extraosseous tumor resistance to αPD-L1 treatment.

Blocking osteoclastogenesis improves extraosseous tumor response to ICB in patients with bone metastasis

RANK-RANKL signaling is essential for osteoclastogenesis.[48,49] Therefore, we evaluated a therapy strategy combining RANKL blocking and αPD-L1 treatment for mice with subcutaneous tumors and bone lesions (Figure S7A). The application of αRANKL is able to sensitize ICB responses in subcutaneous tumors in the presence of bone lesions (Figure S7B). When B16F10-OVA cells were intravenously transferred, the same strategy also effectively reduced lung nodule formation (Figures S7C and S7D). At the molecular level, in mice treated with αRANKL, either as monotherapy or in combination with aPD-L1, we observed that OPN concentrations are significantly reduced in sera, in accordance with our proposed mode of action (Figure S7E). Additionally, we investigated whether zoledronic acid (ZA), the standard-of-care for treating patients with bone metastases in the clinic, could also reduce serum OPN concentration and restore aPD-L1 responsiveness in extraosseous tumors. In contrast to αRANKL, ZA was unable to effectively reduce serum OPN levels or improve the extraosseous response to aPD-L1 treatment in SC + IT tumor-bearing mice (Figures S7F–S7H).

A significant portion of ICB-treated cancer patients are also treated with denosumab, a clinically approved RANKL-neutralizing antibody. In our retrospective cohort AMU8 (Table S1, pan-cancer, $n = 26$), a pan-cancer cohort of bone-metastatic patients under ICB therapies, patients with concurrent treatment with denosumab and ICB showed a significant reduction in extraosseous tumor burden, while bisphosphonate failed to deliver such efficacy (Figures 7A and 7B). The efficacy of denosumab+ICB regimen was confirmed in a similar bone-metastatic pan-cancer cohort, AMU9 (pan-cancer, $n = 150$), with a larger sample size (Figures 7C and 7D). This enhanced anti-tumor responses indeed delivered significant clinical benefit: the PFS in patients treated with the denosumab+ICB regimen was doubled (Figure 7E). Multivariate analysis identified the denosumab-containing regimen as an independent predictor of favorable PFS in patients with bone metastases (Figure S7I).

In a specific case from a patient with metastatic nasopharyngeal carcinoma, after switching the regimen to denosumab plus αPD-1, extraosseous lesions stabilized after four rounds of treatment, and a 50.4% tumor reduction was achieved after eight cycles (Figure 7F), supporting a positive association between RANKL inhibition and improving ICB efficacy in patients with bone metastasis.

We also investigated the correlation between bone metastases and levels of OPN in the sera, as well as its clinical relevance. While established tumors generally elevated OPN levels in sera, patients with bone metastases had significantly higher levels (cohort AMU10, pan-cancer, $n = 83$, Figure 7G). Moreover, stratified analysis of cohort AMU11 (pan-cancer, $n = 34$) indicated that increased OPN concentration during ICB treatments was associated with reduced extraosseous tumor shrinkage (Figure 7H) and poorer overall prognosis, measured by PFS

(L) Growth curves of s.c. tumors in SC + IT mice treated with αPD-L1 and αOPN neutralizing antibody ($n = 5$ for combo group, and $n = 6$ for every other group).
(M) Schematic for tumor implantation and αPD-L1 treatment in *Ctsk*^{Spp1} mice ($n = 6$ for each group).
(N) OPN concentration in serum from *Spp1*$^{f/f}$ ($n = 8$) or *Ctsk*^{Spp1} ($n = 7$) SC + IT mice. Mean ± SEM.
(O) Growth curves of s.c. tumors in SC + IT mice treated with either αPD-L1 or isotype ($n = 4$ for *Ctsk*^{Spp1}+isotype group, and $n = 5$ for every other group).
(P) Survival curves of αPD-L1-treated or control-treated *Spp1*$^{f/f}$ or *Ctsk*^{Spp1} SC + IT mice ($n = 7$ for each group). *p*-value of log rank test are shown.
(Q and R) CD8$^+$ T cells phenotype prior to ICB treatment (10 days after implantation) in subcutaneous tumor by FACS. Red line represents mean value. (Q) Percent in CD45$^+$ immune cell; (R) Number per gram tumors.
(S) Annotation of sequenced CD8$^+$ T cells.
(T) Unsupervised clustering and distribution of CD8$^+$ T cells in s.c. tumor from *Spp1*$^{f/f}$ or *Ctsk*^{Spp1} SC + IT mice.
(U and V) Percent of (U) Exhausted T cells (T_{ex}) and (V) progenitor exhausted T cells (T_{pex}) cells in CD8$^+$ T cells. Bar, mean percentage.
(W) Ratio of TCR clonal number of T_{pex} to T_{ex} cells, stratified according to *Spp1* depletion. Red line, median ratio.
(X) Cumulative frequency expanded T_{pex} clones. Blue bars, highly expanded T cell clones. Statistical analysis: Two-tailed Student's *t* test (D-G, N, Q, R), Mann-Whitney *U*-test (W), two-way ANOVA with Tukey's post hoc test (J, L, O), and log rank test (P) was used. *p*-values <0.05 were considered significant. See also Figures S5 and S6.

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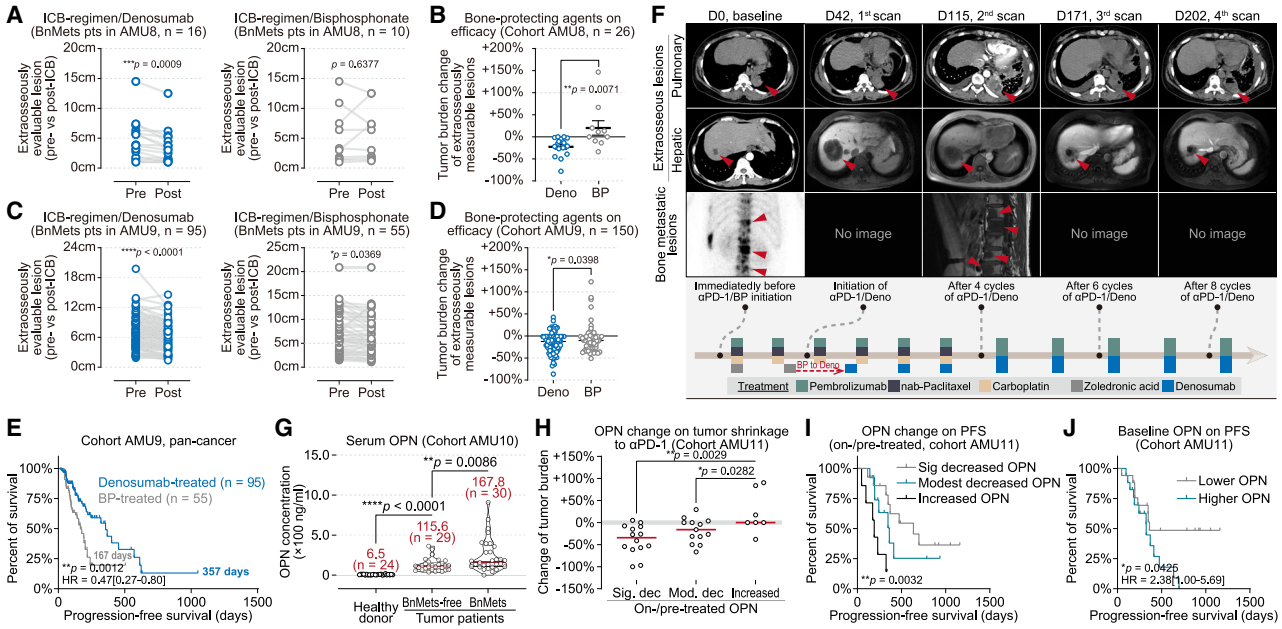


Figure 7. Denosumab+ICB combinational therapy in patients with bone metastasis

(A and C) Extraosseous tumor burden before and after ICB administration, in patients with bone metastasis treated with denosumab or bisphosphonate in cohort (A) AMU8 (n = 26) and (C) AMU9 (n = 150). Two-tailed paired Student's t test.

(B and D) Extraosseous tumor burden change between patients with bone metastasis and treated with denosumab (Deno) or bisphosphonate (BP) in cohort (B) AMU8 and (D) AMU9. Mean ± SEM. Two-tailed Student's t test.

(E) PFS outcomes in response to ICB immunotherapy in bone-metastatic patients treated with either denosumab or bisphosphonate. The hazard ratio with 95% confidential interval, p-value of log rank test are shown.

(F) Clinical course of representative case.

(G) OPN concentration in peripheral blood samples from age-matched healthy volunteers, patients with BnMets disease, and patients with BnMets-free disease in cohort AMU10 (n = 83). Mann-Whitney U-test was used.

(H) Tumor burden changes stratified based on the ratio of post-treatment/pre-treatment OPN concentration. Stratified based on > 100% (increased), <50% (significantly decreased), or otherwise (modestly decreased). Red line, mean tumor burden change. Two-tailed Student's t test was used.

(I) Survival curves of patients with metastatic solid tumors stratified based on change of serum OPN concentrations. Log rank test was used.

(J) Survival curves of patients with metastatic solid tumors stratified based on baseline serum OPN concentrations. Log rank test was used. Statistical analysis: Paired (A, C) and unpaired (B, D, H) two-tailed Student's t -test, Mann-Whitney U-test (G), and log rank test (E, I, J) was used. p-values <0.05 were considered significant. See also Figure S7.

(Figure 7I). Furthermore, a high baseline level of OPN in patient sera itself predicted unfavorable PFS outcomes in ICB immunotherapy (Figure 7J). These results support the serum concentration of OPN as a mechanism-driven biomarker for the prognosis of ICB treatment in patients with bone metastasis.

DISCUSSION

While the link between metastatic site and cancer prognosis is well-established,[10,14,50,51] the organs of metastasis have not been fully incorporated into therapeutic paradigms. Effective immune responses depend on the communication between cells and organs. Disruptions in this communication can enable immune escape, which is a significant challenge in immunotherapy.[24,52,53] As cancer is a systemic disease, we are obligated to contemplate anti-tumor immunity in the whole body: tumor metastasis could alter the immunoregulatory function of recipient organs and therefore reshape the body's immunity systemically.

We found that metastasis in the liver and bone, both immune system-related organs, significantly affected clinical outcomes. It has been elegantly demonstrated that metastases in the liver extensively reprogram anti-tumor responses by enlisting macrophages for tumor antigen-specific T cell elimination,[26] and through differentiation of tumor antigen-specific regulatory T cells.[54] Although similarly impactful to ICB resistance of primary tumors, metastases in the bone utilize a distinct mechanism to promote remote immunosuppression. This suppression is not restricted to the existing anti-tumor T cell immunity; rather, it reprograms the TIME of primary tumors remotely by tumor-conditioned osteoclasts. The activation of osteoclasts for the remolding the bone tissue is necessary to accommodate metastatic tumors in the bone.[55] Metastatic tumors further exploit osteoclast activation to generate a primitive mechanism to suppress immune attacks in extraosseous tumors. Meanwhile, this suppression is less dependent on tumor antigen specificity, unlike liver metastasis which can only inhibit antigen-specific tumor killing.

Disruption of the osteo-immune system can affect extraosseous biology, including muscle, metabolism, and neurocognitive functions. Bones host within them the site for hematopoiesis and for the origin of most immune cells, and recent studies have

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implicated that bone structural cells are integral parts of immunoregulation. It has been shown that ablation of osteocytes leads to severe lymphopenia[56]; osteoblasts supply the differentiation of Siglec-F^{high} neutrophil to facilitate lung tumor growth[52]; and apoptotic bodies generated by the physiological turnover of osteoclasts impair local naive T cell priming, a mechanism favoring bone metastasis of breast cancer.[57] All these bone niche-dependent regulations have systemic implications for anti-tumor immunity. What we discovered here through unbiased proteomics and single-cell transcriptomics approaches is a more classical way of communicating—local tumor-conditioned osteoclasts send protein messengers, OPN, through circulation.

OPN is a multifunctional immunoregulator secreted by a variety of cellular sources, including T cells, NK cells, macrophages, tumor cells, and osteoblasts.[45] Its intracellular and secreted isoforms have distinct functions in hematopoietic cell development[58] and T cell differentiation.[59] The secreted form of OPN, cytokine-like, reduces the progression of acute graft versus host disease[60] and promotes tolerogenic macrophage differentiation and tumor growth.[61] In our animal models bearing bone lesions, OPN cytokine, ~70% of which is produced by tumor-conditioned osteoclasts, reprograms the extraosseous TIME to suppress T cell cytotoxicity. Specifically, the depletion of CD8$^+$ T_{pex} and accelerated exhaustion were observed in the majority of TCR clonotypes, likely impairing T cell differentiation across a large antigen affinity range. While both high- and moderate-affinity TCRs contribute to acute ICB responses, lower-affinity TCRs become dominant for memory formation and sustained ICB efficacy.[62] This implies that OPN may contribute to both primary and acquired ICB resistance in patients with bone metastasis.

To overcome ICB resistance in patients with bone metastasis, our findings strongly suggest that OPN-producing osteoclasts are actionable targets. In clinical applications, we favor αRANKL because it targets upstream osteoclast maturation. Besides blocking the major source of OPN, it also provides benefits for bone protection. Indeed, beside our cohorts, accumulating evidence has shown that αRANKL treatments improve the efficacy of ICB.[63–65] Importantly, although both αRANKL (e.g., denosumab) and bisphosphonates (e.g., zoledronic acid) can treat bone damage, the αRANKL+ICB regimen demonstrated superior efficacy in our cohort compared to the bisphosphonate+ICB combination. This suggests that the immunoregulatory role of αRANKL, mostly through reducing serum OPN, is essential for therapeutic efficacy. Conversely, in the phase III SPLENDOUR trial, denosumab failed to deliver additional clinical benefits over zoledronic acid in NSCLC patients receiving standard platinum-based doublet chemotherapy,[66] strongly supporting an immunoregulatory mechanism underlying the synergistic effects of denosumab+ICB treatment. Our studies urge for more prospective trials to define an optimal denosumab+ICB combinatorial regimen which will improve the clinical benefits of ICB treatment in patients with bone metastasis.

Limitations of the study

There are several limitations to this study. First, our bone metastasis model was established through direct injection rather than hematogenous transfer. To further validate our findings, the development of a spontaneous bone metastasis model that more closely mimics physiological conditions will be necessary in the future. Furthermore, while TAMs and MDSCs have been well-established as significant sources of OPN within the TME, the extent to which tumor-associated osteoclasts contribute to the local microenvironment of extraosseous tumors remains to be elucidated. More importantly, the mechanism by which bone metastases regulate OPN secretion by osteoclasts remains unclear. Additionally, all clinical analyses were retrospective and based on baseline documentation, highlighting the need for prospective clinical trials to better understand the roles of bone metastases in ICB resistance.

RESOURCE AVAILABILITY

Lead contact

Further information and reasonable requests for reagents may be directed to the lead contact, Professor Bo Zhu (bo.zhu@tmmu.edu.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

The raw FASTQ files of RNA-seq in this study can be obtained from Genome Sequence Archive (GSA) with an accession number of CRA012873. The raw data of Proteomics and CyTOF has been deposited on Zenodo with an identifier of https://doi.org/10.5281/zenodo.8412402. All other published datasets used in this study are listed in the key resources table.

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AUTHOR CONTRIBUTIONS

Conceptualization, J.-N.C., Q.J., and B.Z.; clinical investigation, C.S., T.J., X.Z., F.D., Q.Z., and T.Y.; bioinformatic analysis, Z.J. and H.C.; mouse experiment and analysis, J.-N.C., X.Z., J.G., X.L., J.J., Q.Z., X.D., Y.Z., S.X., and X.Y.; intellectual guidance, X.C., H.Z., D.C., Y.Y.W., X.L., L.Y., H.T., and A.L.J.S.; supervision, Q.-J.L., Q.J., and B.Z.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

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Supplemental information can be found online at https://doi.org/10.1016/j.ccell.2025.03.036.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Flow: anti-mouse CD16/CD32, clone 2.4G2	BD Bioscience	Cat# 553142; RRID: AB_394656
Flow: Mouse Lineage Antibody Cocktail	BD Bioscience	Cat# 561317; RRID: AB_10612020
Flow: Mouse Osteopontin/OPN PE-conjugated Antibody	R&D Systems	Cat# IC808P
Flow: CD44 Monoclonal Antibody, clone IM7	eBioscience	Cat# 48-0441-82; RRID: AB_1272246
Flow: anti-mouse CD45, clone 30-F11	BioLegend	Cat# 103137; RRID: AB_2561392
Flow: anti-mouse CD3, clone 17A2	BioLegend	Cat# 100235; RRID: AB_2561455
Flow: anti-mouse CD8, clone 53-6.7	BioLegend	Cat# 100707; RRID: AB_312746
Flow: anti-mouse CD4, clone GK1.5	BioLegend	Cat# 100431; RRID: AB_893329
Flow: anti-mouse CD11b, clone M1/70	BioLegend	Cat# 101229; RRID: AB_2129375
Flow: anti-mouse CD25, clone PC61	BioLegend	Cat# 102012; RRID: AB_312861
Flow: anti-mouse NK1.1, clone PK136	BioLegend	Cat# 108707; RRID: AB_313394
Flow: anti-mouse CD44, clone IM7	BioLegend	Cat# 103005; RRID: AB_312956
Flow: anti-mouse Ki67, clone 16A8	BioLegend	Cat# 652403; RRID: AB_2561524
Flow: anti-mouse F4/80, clone BM8	BioLegend	Cat# 123107; RRID: AB_893500
Flow: anti-mouse Gr-1, clone R86-8C5	BioLegend	Cat# 108411; RRID: AB_313376
Flow: anti-mouse GZMB, clone GB11	BioLegend	Cat# 515408; RRID: AB_2562196
Flow: anti-mouse TNF-α, clone MP6-XT22	BioLegend	Cat# 506321; RRID: AB_961435
Flow: anti-mouse CD45.1, clone A20	BioLegend	Cat# 110729; RRID: AB_1134170
Flow: anti-mouse CD45.2, clone 104	BioLegend	Cat# 109814; RRID: AB_389211
Flow: anti-mouse Tim-3, clone B8.2C12	BioLegend	Cat# 134019; RRID: AB_2814028
Flow: anti-mouse LAG-3, clone C9B7W	BioLegend	Cat# 125225; RRID: AB_2715763
Flow: anti-mouse 2B4, clone m2B4 (B6)458.1	BioLegend	Cat# 133503; RRID: AB_1595624
Flow: anti-mouse Slamf6, clone 330-AJ	BioLegend	Cat# 134609; RRID: AB_2728154
Flow: anti-mouse TIGIT, clone 1G9	BioLegend	Cat# 142109; RRID: AB_2566572
Flow: anti-mouse CD279, clone 29F.1A12	BioLegend	Cat# 135207; RRID: AB_10550092
Flow: anti-mouse IFN-γ, clone XmG1.2	BD Bioscience	Cat# 505825; RRID: AB_1595591
Flow: anti-mouse TCF1, clone S33-966	BD Bioscience	Cat# 566692; RRID: AB_2869822
Flow: anti-mouse FOXP3, clone FJK-16S	Thermo Fisher Scientific	Cat# 25-5773-82; RRID: AB_891552
Flow: anti-mouse TOX, clone TXRX10	Thermo Fisher Scientific	Cat# 50-6502-82; RRID: AB_2574265
Flow: H-2K^{b} p15E Tetramer	MBL	Code No. TS-M507-1
Flow: H-2K^{b} OVA Tetramer	MBL	Code No. TS-5001-1C
Flow: anti-mouse CD366 (Tim-3), clone RMT3-23	BioLegend	Cat# 119715; RRID: AB_2571932
Flow: anti-mouse CD3, clone 17A2	BioLegend	Cat# 100245; RRID: AB_2565882
Flow: anti-mouse Ki-67, clone 11F6	BioLegend	Cat# 151215; RRID: AB_2876504
Flow: anti-mouse CD69, clone H1.2F3	BioLegend	Cat# 104505; RRID: AB_313108
Flow: anti-mouse CD44, clone IM7	BioLegend	Cat# 103037; RRID: AB_10900641
Flow: anti-mouse CD45, clone 30-F11	BioLegend	Cat# 103174; RRID: AB_2860600
Flow: anti-mouse TCR β chain, clone H57-597	BioLegend	Cat# 109218; RRID: AB_493346
Flow: anti-mouse CD279 (PD-1), clone 29F.1A12	BioLegend	Cat# 135220; RRID: AB_2562616
Flow: anti-mouse CD45, clone S18009F	BioLegend	Cat# 157217; RRID: AB_3097472
Flow: anti-mouse CD3, clone 17A2	BioLegend	Cat# 100220; RRID: AB_1732057
Flow: anti-mouse NK-1.1, clone S17016D	BioLegend	Cat# 156506; RRID: AB_2876525

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Flow: anti-mouse CD11c, clone N418	BioLegend	Cat# 117330; RRID: AB_11219593
Flow: anti-mouse CD19, clone 6D5	BioLegend	Cat# 115535; RRID: AB_10933260
Flow: anti-mouse TCR Vβ2, clone B20.6	BioLegend	Cat# 127910; RRID: AB_1089254
Flow: anti-mouse CD28, clone 37.51	BioLegend	Cat# 102108; RRID: AB_312873
Flow: anti-mouse Ki-67, clone 11F6	BioLegend	Cat# 151215; RRID: AB_2876504
Flow: anti-mouse CD366 (Tim-3), clone B8.2C12	BioLegend	Cat# 134010; RRID: AB_2632733
Flow: anti-mouse CD62L, clone MEL-14	BioLegend	Cat# 104424; RRID: AB_493380
Flow: anti-mouse CD3, clone 17A2	BioLegend	Cat# 100236; RRID: AB_2561455
Flow: anti-mouse CD4, clone GK1.5	BioLegend	Cat# 100432; RRID: AB_893323
Flow: anti-mouse FOXP3, clone MF-14	BioLegend	Cat# 126408; RRID: AB_1089115
Flow: anti-mouse F4/80, clone BM8	BioLegend	Cat# 123108; RRID: AB_893500
Flow: anti-mouse Ly-6G, clone 1A8	BioLegend	Cat# 127614; RRID: AB_1877163
Flow: anti-mouse Ly-6C, clone HK1.4	BioLegend	Cat# 128014; RRID: AB_1732079
Flow: anti-mouse CD31, clone 390	BioLegend	Cat# 102418; RRID: AB_830756
Flow: anti-mouse I-A/I-E, clone M5/114.15.2	BioLegend	Cat# 107618; RRID: AB_493525
Flow: anti-mouse CD170 (Siglec-F), clone S17007L	BioLegend	Cat# 155506; RRID: AB_2750234
Flow: anti-mouse CD117 (c-kit), clone ACK2	BioLegend	Cat# 135115; RRID: AB_2561633
Flow: anti-mouse Ly-6A/E (Sca-1), clone D7	BioLegend	Cat# 108120; RRID: AB_493273
Flow: anti-mouse CD34, clone SA376A4	BioLegend	Cat# 152210; RRID: AB_2734219
Flow: anti-mouse CD16/32, clone S17011E	BioLegend	Cat# 156620; RRID: AB_2922485
Flow: anti-mouse CD41, clone MWReg30	BioLegend	Cat# 133923; RRID: AB_2564013
Flow: anti-mouse CD71, clone RI7217	BioLegend	Cat# 113812; RRID: AB_2203383
Flow: anti-mouse TER-119/Erythroid Cells, clone TER-119	BioLegend	Cat# 116212; RRID: AB_313712
Flow: anti-mouse CD127 (IL-7Rα), clone A7R34	BioLegend	Cat# 135010; RRID: AB_1937251
CyTOF: anti-mouse CD45 (clone 30-F11), 89-Y	BioLegend	Cat# 103102; RRID: AB_312967
CyTOF: anti-mouse CD3ε (clone 145-2C11), 115-In	BioLegend	Cat# 100302; RRID: AB_312667
CyTOF: anti-mouse CD28 (clone 37.51), 141-Pr	BioLegend	Cat# 102102; RRID: AB_312867
CyTOF: anti-mouse CD95 (Fas) (clone SA367H8), 142-Nd	BioLegend	Cat# 152602; RRID: AB_2629777
CyTOF: anti-mouse MHC II (I-A/I-E) (clone M5/114.15.2), 143-Nd	BioLegend	Cat# 107602; RRID: AB_313317
CyTOF: anti-mouse CD44 (clone IM7), 144-Nd	BioLegend	Cat# 103002; RRID: AB_312953
CyTOF: anti-mouse CD69 (clone H1.2F3), 145-Nd	BioLegend	Cat# 104502; RRID: AB_313105
CyTOF: anti-mouse CD206 (MMR) (clone C068C2), 146-Nd	BioLegend	Cat# 141702; RRID: AB_10900233
CyTOF: anti-mouse Ly-6G (clone 1A8), 147-Sm	BioLegend	Cat# 127602; RRID: AB_1089180
CyTOF: anti-mouse Ly-6C (clone HK1.4), 148-Nd	BioLegend	Cat# 128002; RRID: AB_1134214
CyTOF: anti-mouse CD19 (clone 6D5), 149-Sm	BioLegend	Cat# 115502; RRID: AB_313637
CyTOF: anti-mouse CD186 (CXCR6) (clone SA051D1), 150-Nd	BioLegend	Cat# 151102; RRID: AB_2566544
CyTOF: anti-mouse CD161(NK-1.1) (clone PK136), 151-Eu	BioLegend	Cat# 108702; RRID: AB_313389
CyTOF: anti-mouse IFN-γ (clone XMG1.2), 152-Sm	BioXCell	Cat# BE0055; RRID: AB_1107694
CyTOF: anti-mouse iNOS (clone CXNFT), 153-Eu	eBioscience	Cat# 14-5920-82; RRID: AB_2572890
CyTOF: anti-mouse CD194 (CCR4) (clone 2G12), 154-Sm	BioLegend	Cat# 131202; RRID: AB_1227524
CyTOF: anti-mouse CD103 (clone 2E7), 155-Gd	BioLegend	Cat# 121402; RRID: AB_535945
CyTOF: anti-mouse CD62L (clone MEL-14), 156-Gd	BioLegend	Cat# 104402; RRID: AB_313089

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
CyTOF: anti-mouse CD25 (clone 3C7), 157-Gd	BioLegend	Cat# 101902; RRID: AB_312845
CyTOF: anti-mouse TIGIT (VSTM3) (clone 2190A), 158-Gd	R&D Systems	Cat# MAB72671
CyTOF: anti-mouse F4/80 (clone Cl: A3-1), 159-Tb	Bio-rad	Cat# MCA497G; RRID: AB_872005
CyTOF: anti-mouse TCR β chain (clone H57-597), 160-Gd	BioLegend	Cat# 109202; RRID: AB_313425
CyTOF: anti-mouse CD64 (FcγRI) (clone X54-5/7.1), 161-Dy	BioLegend	Cat# 139302; RRID: AB_10613107
CyTOF: anti-mouse FOXP3 (clone FJK-16s), 162-Dy	eBioscience	Cat# 14-5773-82; RRID: AB_467576
CyTOF: anti-mouse CD172a (SIRPα) (clone P84), 163-Dy	BioLegend	Cat# 144002; RRID: AB_11203711
CyTOF: anti-mouse CD11c (clone N418), 164-Dy	BioLegend	Cat# 117302; RRID: AB_313771
CyTOF: anti-mouse TCF7(TCF1) (clone 812145), 165-Ho	R&D Systems	Cat# MAB8224
CyTOF: anti-mouse CD27 (clone LG.3A10), 166-Er	BioLegend	Cat# 124202; RRID: AB_1236456
CyTOF: anti-mouse CD184 (CXCR4) (clone L276F12), 167-Er	BioLegend	Cat# 146502; RRID: AB_2562589
CyTOF: anti-mouse CD49b (clone DX5), 168-Er	BioLegend	Cat# 108902; RRID: AB_313409
CyTOF: anti-mouse Ki-67 (clone SolA15), 169-Tm	eBioscience	Cat# 14-5698-82; RRID: AB_10854564
CyTOF: anti-mouse T-bet (clone 4B10), 170-Er	BioLegend	Cat# 644802; RRID: AB_1595503
CyTOF: anti-mouse CD279 (PD-1) (clone 29F.1A12), 171-Yb	BioLegend	Cat# 135202; RRID: AB_1877121
CyTOF: anti-mouse CD127 (IL-7Rα) (clone A7R34), 172-Yb	BioLegend	Cat# 135002; RRID: AB_1937287
CyTOF: anti-mouse CD223 (LAG-3) (clone C9B7W), 173-Yb	BioLegend	Cat# 125202; RRID: AB_961187
CyTOF: anti-mouse CD196 (CCR6) (clone 29-2L17), 174-Yb	BioLegend	Cat# 129802; RRID: AB_1227495
CyTOF: anti-mouse Siglec-F (clone E50-2440), 175-Lu	BD Bioscience	Cat# 552125; RRID: AB_394340
CyTOF: anti-mouse CD366 (Tim-3) (clone RMT3-23), 176-Yb	BioLegend	Cat# 119702; RRID: AB_345376
CyTOF: anti-mouse CD4 (clone RM4-5), 197-Au	BioLegend	Cat# 100576; RRID: AB_2832266
CyTOF: anti-mouse CD8a (clone 53-6.7), 198-Pt	BioLegend	Cat# 100746; RRID: AB_11147171
CyTOF: anti-mouse CD11b (clone M1/70), 209-Bi	BioLegend	Cat# 101202; RRID: AB_312785
In vivo: anti-mouse PD-L1, clone 10F.9G2	BioXCell	Cat# BP0101; RRID: AB_10949073
In vivo: rat IgG2b isotype, clone LTF-2	BioXCell	Cat# BP0090; RRID: AB_1107780
In vivo: anti-mouse RANKL, clone IK22/5	BioXCell	Cat# BE0191; RRID: AB_10949003
In vivo: rat IgG2a isotype, clone 2A3	BioXCell	Cat# BE0089; RRID: AB_1107769
In vivo: anti-mouse osteopontin, clone 103D6	BioXCell	Cat# BE0373; RRID: AB_2927510
In vivo: mouse IgG2c isotype, clone DV5-1	BioXCell	Cat# BE0366; RRID: AB_2894738
Chemicals, peptides, and recombinant proteins		
D-luciferin	Selleck	Cat# S6580
OsteoSense 750 EX	PerkinElmer	Cat# NEV10053EX
Recombinant Mouse M-CSF Protein	R&D Systems	Cat# 416-ML-010/CF
Recombinant Mouse TRANCE/RANK L/TNFSF11 Protein	R&D Systems	Cat# 462-TR-010/CF

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Zoledronic acid monohydrate	Selleck	Cat# S5244
Collagenase IV	Sigma-Aldrich	Cat# 9001-12-1
DNase I	Sigma-Aldrich	Cat# 10104159001
Dispase®II	Sigma-Aldrich	Cat# D4693
Hyaluronidase	Sigma-Aldrich	Cat# H3506
Cell Activation Cocktail	BioLegend	Cat# 423304
Fix/Perm solution	BD Bioscience	Cat# 554722
Fixable Viability Dye eFluor 780	Thermo Fisher Scientific	Cat# 65-0865
CellTrace Violet	Thermo Fisher Scientific	Cat# C34557
CountBright Plus Absolute Beads	Thermo Fisher Scientific	Cat# C36995
Critical commercial assays		
K-ASSAY TRAP staining kit	Kamiya Biomedical Company	Cat# KT-008
APC Annexin V Apoptosis Detection Kit with 7-AAD	BioLegend	Cat# 640930
Mouse RANKL ELISA kit	R&D Systems	Cat# MTR00
Human RANKL ELISA kit	Elabscience	Cat# E-EL-H5813c
Mouse OPN ELISA kit	R&D Systems	Cat# MOST00
Human OPN ELISA kit	R&D Systems	Cat# DOST00
Mouse IGF-1 ELISA Kit	Invitrogen	Cat# EMIGF1
Mouse Lactoferrin ELISA Kit	Invitrogen	Cat# EEL103
Mouse FBN1/Fibrillin1 ELISA Kit	LSBio	Cat# LS-F24166
Mouse CLIC1/NCC27 ELISA Kit	LSBio	Cat# LS-F25367
Mouse RHOA ELISA Kit	LSBio	Cat# LS-F38474
EasySep™ Mouse $CD8^+$ T Cell Enrichment Kit	STEMCELL	Cat# 19753
Deposited data		
RNA-Seq	This study	GSA: CRA012873
Proteomics and CyTOF	This study	Zenodo: https://doi.org/10.5281/zenodo.8412402
Clinical dataset of melanoma treated with PD-1 blockade	Liu et al.[27]	Supplementary Tables
Experimental models: Cell lines		
B16F10	Chinese Academy of Medical Sciences	1101MOU-PUMC000473
MC38	Chinese Academy of Medical Sciences	1101MOU-PUMC000523
CT26	Chinese Academy of Medical Sciences	1101MOU-PUMC000275
E.G7-OVA	TMMU	N/A
Experimental models: Organisms/strains		
C57BL6/J	TMMU	N/A
BALB/c	TMMU	N/A
Ctsk-2A-Cre mice	Shanghai Model Organisms Center, Inc.	Cat# NO. NM-KI-19001
Spp1-Flox mice	Shanghai Model Organisms Center, Inc.	Cat# NO. NM-CKO-210205
Oligonucleotides		
Actin primer (mouse): Fw: ATGGAGCCGGACAGAAAAGC; Rv: CTTGCCACTCAGGGAAGGA	Beijing Tsingke Biotech Co., Ltd.	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Spp1 primer (mouse): Fw: AGCAAGAAACTCTTCCAAGCAA; Rv: GTGAGATTCGTCAGATTCATCCG	Beijing Tsingke Biotech Co., Ltd.	N/A
Recombinant DNA		
CV146-OVAL	Genechem	N/A
CV146-Luciferase	Genechem	N/A
Software and algorithms		
GraphPad Prism	GraphPad Software	www.graphpad.com
FlowJo Software (verison 10.9.0)	TreeStar	www.flowjo.com
Living Image	PerkinElmer	www.perkinelmer.com
Proteome Discover-2.4	Thermo Fisher Scentific	https://www.thermofisher.com
R-4.2.1	Team, R.C.[67]	https://www.r-project.org
fgsea-1.18	Korotkevich et al.[68]	https://github.com/ctlab/fgsea/
Hisat2-2.2.0	Kim et al.[69]	http://daehwankimlab.github.io/hisat2/
featureCounts-2.0.1	Liao et al.[70]	https://subread.sourceforge.net/featureCounts.html
DESeq2-1.32.0	Love et al.[71]	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
clusterProfiler-4.8.2	Yu et al.[72]	https://github.com/YuLab-SMU/clusterProfiler
CATALYST-1.20.1	Chevrier et al.[73]; Nowicka et al.[74]	https://github.com/HelenaLC/CATALYST
cytofkit-1.10.0	Chen et al.[75]	https://github.com/JinmiaoChenLab/cytofkit
ggpointdensity-0.1.0	Kremer[76]	https://github.com/LKremer/ggpointdensity
ggplot2-3.4.3	Wickham[77]	https://github.com/tidyverse/ggplot2
factoextra-1.0.7	Kassambara & Mundt[78]	http://www.sthda.com/english/rpkgs/factoextra
Cell Ranger-7.1.0	10X Genomics	https://www.10xgenomics.com
Seurat-4.3.0	Hao et al.[79]	https://satijalab.org/seurat/
scRepertoire-2.0.0	Borcherding et al.[80]	https://github.com/ncborcherding/scRepertoire
randomForest-4.6-14	10.32614/CRAN.package.randomForest	https://cran.r-project.org/web/packages/randomForest/index.html
tidyverse-2.0.0	https://doi.org/10.32614/CRAN.package.tidyverse	https://cran.r-project.org/web/packages/tidyverse/index.html

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cohort study

Patients in cohorts AMU1, AMU2, AMU3, AMU4, AMU5, AMU6, AMU7, AMU8, AMU9, AMU10, and AMU11 were recruited through the Institute of Cancer, Xinqiao Hospital, Third Military Medical University. Patients in cohorts TUSM1, TUSM2, and TUSM3 were recruited through the Department of Medical Oncology, Shanghai Pulmonary Hospital, Tongji University School of Medicine. Cohort identifier, number of cases, tumor type, intervention, related figures, and available clinical information were summarized in Table S1. Clinical characteristics and radiological following-ups were retrospectively obtained with the approval of Institutional Review Broad. For publicly available cohort, Nat Med 2019, melanoma,[27] all relevant clinical documents were downloaded from supplementary material online.

Cohort AMU1 represents patients with metastatic solid tumors who received ICB-containing regimen at 1st ward in Institute of Cancer, Xinqiao Hospital from Aug 2017 to Jul 2023. Cohort AMU2 represents patients with metastatic solid tumors who received ICB-containing regimen at 2nd ward in Institute of Cancer, Xinqiao Hospital from Aug 2017 to Jul 2023. Cohort AMU3 represent patients with metastatic solid tumors who received chemotherapy (either chemotherapy alone or combining with radiotherapy, anti-angiogenic agents, anti-Her2 treatment, or endocrine therapy) at 2nd and 3rd wards in Institute of Cancer, Xinqiao Hospital from Jan

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2015 to Dec 2018. Patients receiving EGFR/ALK/ROS1/RET target therapy were excluded from cohort AMU3. Cohort AMU4 represents patients with metastatic NSCLC who received chemotherapy (either chemotherapy alone or combining with radiotherapy or anti-angiogenic agents) at 1st ward in Institute of Cancer, Xinqiao Hospital from Jan 2015 to Dec 2018. Patients receiving EGFR/ALK/ROS1/RET target therapy were excluded from cohort AMU4. Cohort AMU5 represents patients with extended stage small cell lung cancer who received chemotherapy (either chemotherapy alone or combining with radiotherapy) at 1st ward in Institute of Cancer, Xinqiao Hospital from Jan 2015 to Dec 2018. Cohort AMU6 and AMU7 represent patients with metastatic solid tumors who received ICB-containing regimen at Institute of Cancer, Xinqiao Hospital from Jul 2018 to Aug 2022, and from Sep 2022 to Aug 2023, respectively. Sum of diameters of extraosseous measurable lesions in pre- (baseline) and post-ICB-treated (radiological scans after 2∼4 cycles of ICB-containing treatment) radiological imaging were recorded for patients in cohort AMU6 and AMU7. Cohort AMU8 and AMU9 represents patients with bone-metastatic solid tumors who received ICB-containing regimen, either combining with denosumab or bisphosphonate/zoledronic acid therapy, at Institute of Cancer, Xinqiao Hospital from Jul 2018 to Aug 2022, and from Sep 2022 to Aug 2023, respectively. Sum of diameters of extraosseous measurable lesions in pre- (baseline) and post-ICB-treated (radiological scans after 2∼4 cycles of ICB-containing treatment) radiological imaging were recorded for patients in cohort AMU8 and AMU9. Cohort AMU10 represent healthy volunteers, treatment-naive cancer patients with bone metastasis or not. Peripheral blood samples were collected from patients in cohort AMU10 detect OPN concentration. Cohort AMU11 were solid cancer patients with bone-metastasis who received ICB-containing treatment. Sum of diameters of extraosseous measurable lesions in pre- (baseline) and post-ICB-treated (radiological scans after 2∼4 cycles of ICB-containing treatment) radiological imaging and PFS were recorded.

Cohort TUSM1 represents patients with metastatic NSCLC who received ICB-containing regimen at Department of Medical Oncology, Shanghai Pulmonary Hospital, Tongji University School of Medicine from Jan 2018 to Dec 2020. Cohort TUSM2 represents patients with metastatic NSCLC who received ICB-containing regimen at Department of Medical Oncology, Shanghai Pulmonary Hospital, Tongji University School of Medicine from Jan 2018 to Dec 2020. All patients were liver-metastatic and brain-metastasis-free in cohort TUSM2. Cohort TUSM3 represents patients with metastatic NSCLC who received ICB monotherapy at Department of Medical Oncology, Shanghai Pulmonary Hospital, Tongji University School of Medicine from Jan 2018 to Dec 2020.

In each participating site, two associate chief physicians or senior attending physicians reviewed the radiological archives to define the presence of measurable lesions and evaluate the diameter of each measurable lesion. Objective response rate was determined according to RECIST 1.1 criteria. PFS and OS were calculated from the initial administration of ICB agents.

Animal studies

Six-to-eight-week-old male C57BL/6 and BALB/c mice were ordered from the Center of Experimental Animals of Third Military Medical University (TMMU). Mice were housed under standard pathogen free conditions. Ctsk-2A-Cre mice (Cat. NO. NM-KI-190019) and *Spp1*-Flox mice (Cat. NO. NM-CKO-210205) were purchased from Shanghai Model Organisms Center, Inc. The *Ctsk*-2A-Cre mice and *Spp1*-Flox mice were crossed to generate *Spp1* conditional knockout ($Ctsk^{Cre}$; $Spp1^{f/f}$) mice. All mice experiments performed at Xinqiao Hospital were following the institutional animal care and Laboratory Animal Welfare and Ethics Committee of TMMU (NO: AMUWEC20210386; animal production license number: SCXK20170002; animal use license number: SYXK20170002).

Cell lines

The mouse colon adenocarcinoma cell line MC38, mouse melanoma cell line B16F10, Lewis lung carcinoma cell line LLC and colon carcinoma cell line CT26 were obtained from the CAMS (Cell Resource Center, Institute of Basic Medicine, Chinese Academy of Medical Sciences). Luciferase-expressing B16F10 tumor cells (B16F10-luciferase), ovalbumin-expressing B16F10, LLC and MC38 tumor cells (B16F10-OVA, LLC-OVA, and MC38-OVA) were established by transfection with lentivirus and selected with puromycin. Cells were cultured in Dulbecco's modified Eagle medium (DMEM) or RPMI-1640 medium supplemented with 10% fetal bovine serum (FBS), penicillin (100 IU/mL), streptomycin (100 μg/mL), 2 mM L-glutamine, and 1 mM sodium pyruvate in a humidified incubator at 37°C and 5% CO_2.

METHOD DETAILS

Tumor models

For subcutaneous tumor models, B16F10-luciferase (1×10^6), B16F10-OVA (1×10^6), B16-F10 tumor cells (1×10^6), LLC-OVA (1×10^6), E.G7-OVA (5×10^5), MC38 tumor cells (5×10^5), and CT26 tumor cells (5×10^5) were suspended in 100 μL cell culture medium and subcutaneously injected on the right hindlimb. For osseous tumor-bearing mouse models, mouse tumor cell line B16F10-luciferase (5×10^5), B16F10-OVA (5×10^5), B16-F10 (5×10^5), LLC-OVA (5×10^5), E.G7-OVA (2.5×10^5), MC38 (2.5×10^5) or CT26 (2.5×10^5) tumor cells were digested and resuspended of 5×10^7/mL cells in cell culture medium. Mice were anesthetized with pentobarbital (80 mg/kg) and a superficial incision about 0.5 cm was made to expose the patellar ligament. Firstly, a 25-gauge needle was used to create a route from the patellar ligament to tibial cavity. Then the needle was replaced with a 10 μL microinjection syringe containing suspension of tumor cells. After slowly injecting tumor cells, the incision was covered with gelatin sponge to prevent liquid leakage. Tumor size was measured every other day and calculated as (length × width^2)/2. Mouse survival was monitored every other day. For lung metastasis model, single-cell suspensions of B16F10-OVA tumor cells (1×10^6) were intravenously injected into the tail vein of mouse. Treatment was started on days 10, and lungs were harvested for counting surface tumor nodules, weighing and H&E staining on days 17.

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Mouse treatment protocol

For immune checkpoint blockade therapy, 200 μg anti-mouse PD-L1 (BioXCell, BP0101) or isotype IgG2b (BioXCell, BP0090) was given intraperitoneally on days 10, 13, 16. For anti-mouse RANKL (BioXCell, BE0191) treatment, mice were given 100 μg antibody or isotype IgG2a (BioXCell, BE0089) by intraperitoneally injection every other day, and four times in total. For zoledronic acid monohydrate (Selleck, S5244) treatment, 100 μg/kg agent was given subcutaneously daily for totally seven times. For anti-mouse osteopontin (BioXCell, BE0373) intervention, 200 μg antibody or isotype IgG2c (BioXCell, BE0366) was given intraperitoneally every three days, and three times in total. For mouse serum transfer experiments, mouse orbital blood from tumor-free mouse, osseous tumor-bearing mouse, or subcutaneous tumor-bearing mouse was collected to separate serum through centrifugation. The number of serum donor mice was the same as that of the recipient mice. The collected serum was mixed thoroughly and then transfused to the subcutaneous tumor-bearing mice by intraperitoneally injection every other day, and three times in total. For osteoclast-condition medium treatment, successfully induced osteoclasts were washed with PBS to exclude cytokines and then cultured with serum-free medium for two days before the conditioned medium was collected for transfer experiment. When tumor-induced osteoclast supernatant transfer, the tumor conditioned medium is incubated with osteoclasts for an additional two days before *in vivo* transfer.

In vivo imaging

For *in vivo* bioluminescence imaging (BLI), B16F10-luciferase tumor cells were used to establish mouse tumor model. Mice were anesthetized with isoflurane (2% in 1 L/min oxygen) and given an intraperitoneal injection of 200 μL D-luciferin (Selleck, 15 mg/mL). BLI images were acquired with the IVIS Spectrum system (PerkinElmer). For Micro-CT, mice were anesthetized with isoflurane and assayed by PerkinElmer Quantum FX system. For OsteoSense signal detection, mouse hair from hind legs and lower abdomen were removed, 4 nmol/100 μL OsteoSense 750 EX (PerkinElmer, NEV10053EX) was injected intravenously into the mouse. The optimal imaging time point is 3-24 hours post injection according to the instructions using an imaging system BLT (AniView100).

Osteoclast induction

Bone marrow from the femur and tibia of six to eight-week-old mouse was flushed out with cold DMEM medium and processed into a single-cell suspension without red blood cells. After cultured in incubator at 37°C for 30 minutes, nonadherent cells were collected through centrifuge at 2200 rpm for 5 minutes. Cells were resuspended in osteoclast induction medium of DMEM complete medium with 50 ng/mL M-CSF (R&D Systems) and 50 ng/mL RANKL (R&D Systems) at a concentration of 1×10^5 cells/mL. The medium was replaced by fresh culture medium every two days, and 7-10 days later osteoclasts can be harvested for detection with K-ASSAY TRAP staining kit (Kamiya Biomedical Company) according to the manufacturer's instructions.

Histological analysis

For histologically analysis of mouse osseous tumor, leg specimens were harvested and fixed with 10% formalin for two days, then the muscle tissues around bone were removed carefully for EDTA decalcification. Check the specimen daily for adequate decalcification solution coverage and change the solution every three days. 14 days later the specimens were embedded in paraffin and cut into 5-micrometer-thick sections for H&E staining. The specimens were scanned by MoticDSM System.

Flow cytometry

Tumor tissues were cut into little specimens and digested with collagenase IV (1 mg/mL, Thermo Fisher Scientific), dispase 2 (0.1 mg/mL, Sigma-Aldrich), hyaluronidase (0.1 mg/mL, Sigma-Aldrich) and deoxyribonuclease (DNase I) (0.2 mg/mL, Sigma-Aldrich). For intracellular staining, lymphocytes were incubated with RPMI-1640 complete medium supplemented with Cell Activation Cocktail (with Brefeldin A, 423304, BioLegend) at 37°C for 4 hours. For extracellular staining, the cells were pre-cultured with CD16/CD32 (clone 2.4G2, BD Bioscience) at 4°C for 30 minutes to block non-antigen specific binding of immunoglobulins. The cells were incubated with extracellular antibodies at 4°C for 30 minutes, then washed and fixed by Fix/Perm solution (BD Bioscience) for 1 hour, and finally intracellular staining for 40 minutes. The following antibodies were used: CD3 (clone 17A2), CD4 (clone GK1.5), CD8α (clone 53-6.7), CD11b (clone M1/70), CD11c (clone N418), CD16/32 (clone S17011E), CD19(clone 6D5), CD25 (clone PC61), CD28 (clone 37.51), CD31 (clone 390), CD34 (clone SA376A4), CD41 (clone MWReg30), CD44 (clone IM7), CD45 (clone 30-F11; clone S18009F), CD45.1 (clone A20), CD45.2 (clone 104), CD62L (clone MEL-14), CD69 (clone H1.2F3), NK1.1 (clone PK136; clone S17016D), CD71 (clone RI7217), CD117 (c-kit)(clone ACK2), CD127 (IL-7Rα)(clone A7R34), CD170 (Siglec-F)(clone S17007L), CD279 (clone 29F.1A12), APC Annexin V Apoptosis Detection Kit with 7-AAD (640930), F4/80 (clone BM8), Gr-1 (clone R86-8C5), GZMB (clone GB11), I-A/I-E (clone M5/114.15.2), Ki67 (clone 16A8; clone 11F6), LAG-3 (clone C9B7W), Ly-6A/E (Sca-1)(clone D7), Ly-6G (clone 1A8), Ly-6C (clone HK1.4), TER-119/Erythroid (clone TER-119), TNF-α (clone MP6-XT22), TCR Vα2 (clone B20.1; clone B20.6), Tim-3 (clone B8.2C12), 2B4 (clone m2B4 (B6)458.1), Slamf6 (clone 330-AJ), TIGIT (clone 1G9), above antibodies were all purchased from BioLegend, and others were detailed mentioned: IFN-γ (clone XmG1.2, BD Biosciences), mouse Lineage Antibody Cocktail (BD Biosciences), TCF-1 (clone S33-966, BD Biosciences), Osteopontin/OPN (clone, R&D Systems), Fixable Viability Dye eFluor 780 (65-0865, Thermo Fisher Scientific), CellTrace Violet (CTV) (Thermo Fisher Scientific, C34557), FOXP3 (clone FJK-16S, Thermo Fisher Scientific), TOX (clone TXRX10, Thermo Fisher Scientific), CountBright Plus Absolute Beads (C36995, Thermo Fisher Scientific), H-2K^b p15E Tetramer (KSPWFTTL, MBL), H-2K^b OVA Tetramer (SIINFEKL, MBL).

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Quantitative proteomic sequencing and data processing

The orbital blood from tumor-free mice, subcutaneous tumor-bearing mice, and osseous tumor-bearing mice were collected respectively after mice were anesthetized with pentobarbital (80 mg/kg), and serum samples were separated by centrifugation at 1000g for 15 minutes at 4°C and stored in aliquots at -80°C. Subsequent project procedures including protein extraction, process, peptide *in vitro* label of Tandem Mass Tag (TMT) technique and Liquid Chromatography-Mass Spectrometry LC-MS/MS analysis were conducted by EVbio (Beijing). Briefly, mouse serum was collected for protein identification using HPLC-MS/MS technique. Sample was dissolved in lysis buffer for protein extraction and digestion, the total protein concentration was measured by the Bradford protein assay. After TMT labeling, the acquired peptide fractions were loaded into the nano UPLC-MS/MS system (Thermo Fisher Scientific). The resulting MS/MS data were quantification and processed using Proteome Discover 2.4.

The original data was filtered with null values and performed median normalization to eliminate the experiment error. The proteins with fold change more than 1.5 or less than 0.66 and p-value less than 0.05 were recognized as differentially expression proteins. For GSEA analysis, fgsea package with Gene Oncology (GO) gene sets were applied.[68]

RNA sequencing and data processing

Mouse tumor tissues and osteoclasts were stored in TRIzol reagent (Thermo Fisher Scientific) and prepared by Novogene Bioinformation Technology Co, Ltd. for transcriptome sequencing (Beijing, China). RNA was extracted from the samples according to standard methods, and total amounts and integrity of RNA were assessed using the RNA Nano 6000 Assay Kit of the Bioanalyzer 2100 system (Agilent Technologies, CA, USA). Total RNA was used as input material for the mRNA purified and subsequent library construction. Raw reads of fastq format were firstly processed to remove reads containing adapter, reads containing poly-N and low-quality reads. All downstream deep analyses were based on the clean data with high quality.

The clean data were further aligned to GRCm38 reference genome using Hisat2.[69] Uniquely mapped reads were counted with featureCounts using Ensembl GRCm38 GTF transcript annotations,[69] and normalized based on the length of the gene and reads count mapped to this gene. To perform the "Immune maps", a random forest machine learning algorithm using 244 input variables.[30] RandomForest R package was used to run this analysis. MDS was used to assign a spatial location to each sample according to the proximity matrix from machine learning and 244 dimensions of defined immune-related features.

To identify the differentially expression genes between conditions, log2 fold change (log2FC) and adjusted *p*-value were calculated through DESeq2.[71] The GSEA function from clusterProfiler package with GO, HALLMARK, REACTOME, and BIOCARTA pathway gene sets of MSigDB were used to perform GSEA analysis based on ranked genes.[72,81,82]

Quantitative RT-PCR

Total RNA was extracted from tumor tissues using the TRIzol reagent (Thermo Fisher Scientific), followed by complementary DNA (cDNA) synthesis using a PrimeScript RT reagent kit according to the manufacturer's protocol (Takara). Gene expression was assessed using a CFX384 Real-Time PCR detection system with Premix Ex Taq II (Takara) and gene-specific primers as shown in key resources table. Cycling conditions included a single denaturing step at 95°C for 30 seconds, followed by 40 cycles at 95°C for 5 seconds and 60°C for 34 seconds.

Mass cytometry sample preparation

The sample preparation was performed according to the manufacture's instruction (PLT-Tech Inc, Hangzhou, China). Briefly, tumor tissues were mechanically dissociated with the GentleMACS Dissociator (Miltenyi Biotec) and digested into single cells with 0.2 mg/mL Deoxyribonuclease (DNase I), Collagenase IV and hyaluronidase (Sigma-Aldrich) for 30 minutes at 37°C. Red cells were removed using ACK Lysing Buffer (PLT-Tech). 3×10^6 single cells were pulsed with 0.25 μM Cell-IDTM Cisplatin-194Pt (Fluidigm, Cat# 201194) and then blocked with anti-CD16/CD32 (clone 2.4G2, BD Bioscience). Qualified cells were stained with forty-one metal-conjugated antibody cocktails (key resources table), followed by fixation with 250 nM Cell-IDTM Intercalator-Ir (Fluidigm, Cat# 201192B) in Fix and Perm Buffer (Thermo Scientific, Cat# 00-5123-43, Cat# 00-5223-56). After incubated with intracellular antibody cocktail, the cells were washed with Maxpar Fix and Perm Buffer (Fluidigm, Cat# 201067) and mixed with EqBeads before analysis on a CyTOF system (Helios, Fluidigm).

Analysis of mass cytometry data

After acquisition, FCS files were generated and imported into FlowJo to manually gate living $CD45^+$ singlets. The data of gating cells were integrated and transformed using arcsinh with cofactor 5 to make the expression value comparable.[83] To perform the similarity of samples, pseudobulk-level Multi-Dimensional Scaling (MDS) plot was calculated based on the median expression of all 41 proteins with pbMDS function from CATALYST package.[73,74] The Rphenograph clustering algorithm,[84] available through the cytofkit package,[75] was applied to generate clusters. Clusters were annotated into 10 major cell types, such as $CD8^+$ T cell, $CD4^+$ T cell, and Macrophage, according to canonical immune cell markers. The annotation data was visualized using *t*-Distributed Stochastic Neighbor Embedding (*t*-SNE) via cytof_dimReduction function through cytofkit package, and density plots via the ggpointdensity package.[76] To investigate domain major features which cause the difference between two conditions, principal component analysis (PCA) was performed on the fraction matrix of major cell type, and the contribution of each major cell type was calculated through factoextra package.[78]

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ELISA

Protein levels in the serum or condition medium were quantified using ELISA kit according to the manufacturer's instructions: Mouse RANKL ELISA kit (MTR00, R&D Systems), Human RANKL ELISA kit (E-EL-H5813c, Elabscience), Mouse OPN ELISA kit (MOST00, R&D Systems), Human OPN ELISA kit (DOST00, R&D Systems), Mouse IGF-1 ELISA Kit (EMIGF1, Invitrogen), Mouse Lactoferrin ELISA Kit (EEL103, Invitrogen), Mouse FBN1/Fibrillin1 ELISA Kit (LS-F24166, LSBio), Mouse CLIC1/NCC27 ELISA Kit (LS-F25367, LSBio), Mouse RHOA ELISA Kit (LS-F38474, LSBio). The concentration of protein was investigated in duplicate and calculated by extrapolating values according to standard curve following manufactures guidelines.

Single-cell RNA-seq library and single-cell TCR-seq library preparation and sequencing

Single-cell libraries were prepared using the Chromium Single Cell 5' Library and Gel Bead Kit, according to the manufacturer's instructions. Briefly, cells were washed with 0.04% BSA DPBS three times and resuspended to a concentration of 700-1200 cells/μL (viability $\geq$ 85%) as determined by Countess® II. Around 10,000 cells (targeted cell recovery rate ~90%) were encapsulated in droplets with barcoded beads and underwent reverse transcription to generate barcoded cDNA. After the RT step, emulsions were broken and barcoded cDNA was purified with Dynabeads, followed by PCR amplification. Amplified cDNA was used for both gene expression (GEX) and TCR/BCR library construction. For GEX libraries, fragmented cDNA (50 ng) underwent end-repair, size selection with SPRIselect beads (optimal size range not specified), and adapter ligation for Illumina sequencing. The final library was sequenced on a NovaSeq platform, generating 150 bp paired-end reads. For TCR/BCR libraries, TCR transcripts were specifically amplified with 10 PCR cycles. The VDJ region-enriched libraries were size-selected with SPRI beads (optimal size range not specified) and sequenced on a NovaSeq 6000 instrument, generating 150 bp paired-end reads.

Single-cell RNA-seq and single-cell TCR-seq data processing

The single-cell RNA and TCR sequencing data from the same sample were processed using the Cell Ranger multi pipeline (version 7.1.0) from 10x Genomics. The scRNA-seq data was aligned to mouse reference genome (GRCm38), assigned cell barcodes, and used to generate the unique molecular identifier (UMI) matrix. A R-based toolkit, Seurat (version 4.3.0),[79] was used for analyzing the scRNA-seq data. Cells with less than 500 UMIs, less than 200 genes detected, and more than 10% mitochondrial gene counts were filtered out. The TCR sequences were aligned to the same mouse reference genome and filtered to retain high-quality, full-length, productive, and assigned with a valid cell barcode and an unambiguous chain type. ScRepertoire (version 2.0.0) was applied to remove chains without values and keep the chain with the highest UMI count if a cell had two or more qualified chains of the same type.[80] The combined TCR information were integrated into the same Seurat object using AddMetaData() function.

Identification of exhaustion progenitor CD8$^+$ T cells

To accurately identify the exhaustion progenitor CD8$^+$ T cells, we employed a two-round analysis strategy. In the first-round, each sample was analyzed independently, for each sample, the top 2,000 highly variable genes (HVGs) were identified using the "vst" method. The normalized data was obtained by the "LogNormalize" method and scaled with the mean and standard deviation of the HVGs. Principal component analysis (PCA) was performed on the variable gene matrix to reduce noise, and the top 30 out of 50 principal components were used for downstream analyses. A K-nearest neighbor (KNN) graph was constructed based on the Euclidean distance in PCA space. The edge weights between any two cells were refined based on the co-expression in their local neighborhoods. The Louvain algorithm was applied to identify cell clusters with a resolution of 0.5. We then used the same principal components for non-linear dimension reduction to generate the Uniform Manifold Approximation and Projection (UMAP) for visualization. Cell clusters with low expression of CD45, CD3E, and CD8A, and/or high expression of ITGAM, were removed. We first annotated the terminally exhaustion CD8$^+$ T cell clusters in each sample based on the high expression of Havcr2, Lag3, Pdcd1, and Entpd1. Exhaustion progenitor CD8$^+$ T cells were then identified as cells sharing the same clonotype with terminally exhaustion cells. Here, the same clonotype refers to identical nucleotide sequences for both α- and β-chain. The remaining cells were annotated as proliferative CD8$^+$ T cells with high expression of Mki67, or non-exhaustion CD8$^+$ T cells.

After the first-round analysis, data from *Spp1*$^{f/f}$ and *Ctsk*^{Spp1} samples were integrated. The merged data were then re-normalized, dimensionally reduced, and visualized using the methods mentioned above. No obvious batch effect was not observed during the second-round integration analysis. The annotation was inherited from the first-round analysis for visualization.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data were analyzed using GraphPad Prism Software (version 8.2.1), SPSS (IBM), and R Studio. Experiments were repeated two to three times, and comparisons between two groups were assessed by a two-tailed Student's *t*-test. Multiple comparisons of tumor sizes were assessed by two-way analysis of variance (ANOVA). HR was determined through the univariable and multivariable Cox regression. Fourteen clinically relevant parameters, encompassing demographic data, treatment history, and the presence of metastases in other organs, were incorporated into the multivariable Cox regression analysis. Kaplan-Meier estimates and log-rank (Mantel-Cox) test were used to analyze survival curves. *P* values <0.05 were considered statistically significant.